# Linear Programming (Lec11.pdf)

Ioannis Akrotirianakis

# Todays topics & Logistics

- Linear Programming (LP)
  - Applications/History
- Geometry of LP
  - Vertex & Extreme points
- The Simplex method
  - Main idea and motivation
  - Examples

- **HW5** → due tonight!
- **HW6** → due April 9 @ 11:59PM

# General formulation of a Linear Program

- The standard form

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

  - Data: $c \in R^n$, $A \in R^{m \times n}$, $b \in R^m$
  - Decision variables: $x \in R^n$
  - Linear objective and affine inequalities and/or equalities

# General formulation of a Linear Program

- Expanded standard form:

$$
\begin{aligned}
min \quad & \sum_{i=1}^{n} c_i x_i \\
s.t. \quad & a_j^T x = b_j, \qquad j = 1, \dots, m \\
& x_i \geq 0, \qquad i = 1, \dots, n
\end{aligned}
$$

$$
A = \begin{pmatrix} \leftarrow a_1^T \rightarrow \\ \dots \\ \leftarrow a_m^T \rightarrow \end{pmatrix} \in R^{m \times n}
$$

- Other forms of an LP

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & A_1 x = b_1 \\ & A_2 x \leq b_2 \\ & A_3 x \geq b_3 \end{aligned}$$

- Data: $c \in R^n$, $A_1 \in R^{m_1 \times n}$, $b_1 \in R^{m_1}$, $A_2 \in R^{m_2 \times n}$, $b_2 \in R^{m_2}$ , $A_3 \in R^{m_3 \times n}$, $b_3 \in R^{m_3}$
- Decision variables: $x \in R^n$
- An LP can always be converted to the standard form by adding slack variables

- Other forms of an LP

What type of Optimization problem is an LP??

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What type of Optimization problem is an LP??

Convex Optimization problem.
We can use CVX solver!

- Data: $c \in R^n$, $A_1 \in R^{m_1 \times n}$, $b_1 \in R^{m_1}$, $A_2 \in R^{m_2 \times n}$, $b_2 \in R^{m_2}$ , $A_3 \in R^{m_3 \times n}$, $b_3 \in R^{m_3}$

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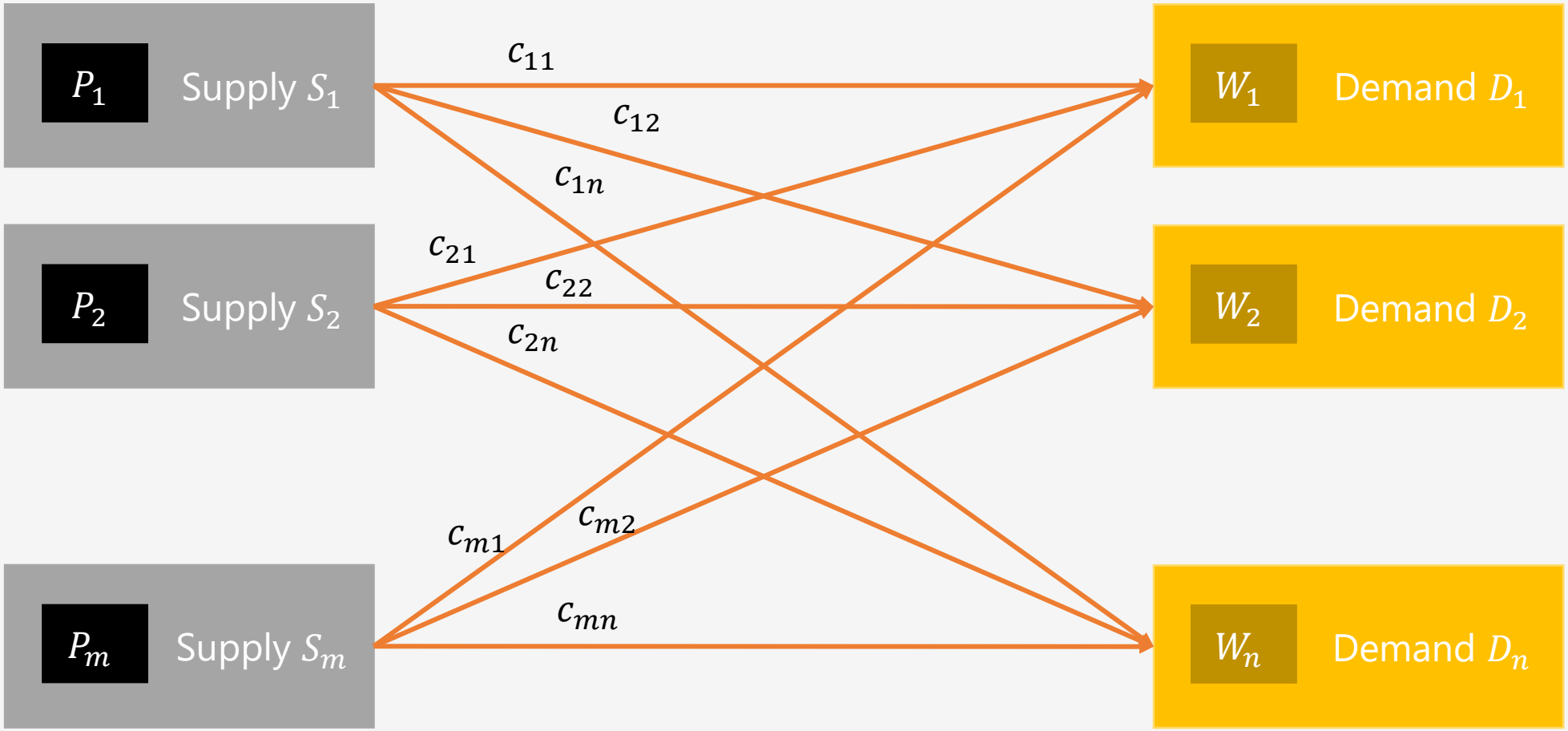
# Examples of LPs: Optimal Transportation Cost Problem



# Examples of LPs: Optimal Transportation Cost Problem

- **Data:** Supply $S_i$, Demand $D_j$, capacities of transportation network $c_{ij}$
- **Decision variables**: the quantity of product produced in plant $i$ that needs to be transported to warehouse $j$
- **Objective**: transport all products produced in Plants $P_1, \dots, P_m$ to Warehouses $W_1, \dots, W_n$ with the <u>minimum cost.</u>
- **Constraints**:
  - must satisfy the demand of each warehouse
  - must not exceed supply capacity of each plant

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- **Data:** Supply $S_i$, Demand $D_j$, capacities of transportation network $c_{ij}$

- **Decision variables**: the quantity of product produced in plant $i$ that needs to be transported to warehouse $j$ : $x_{ij}$

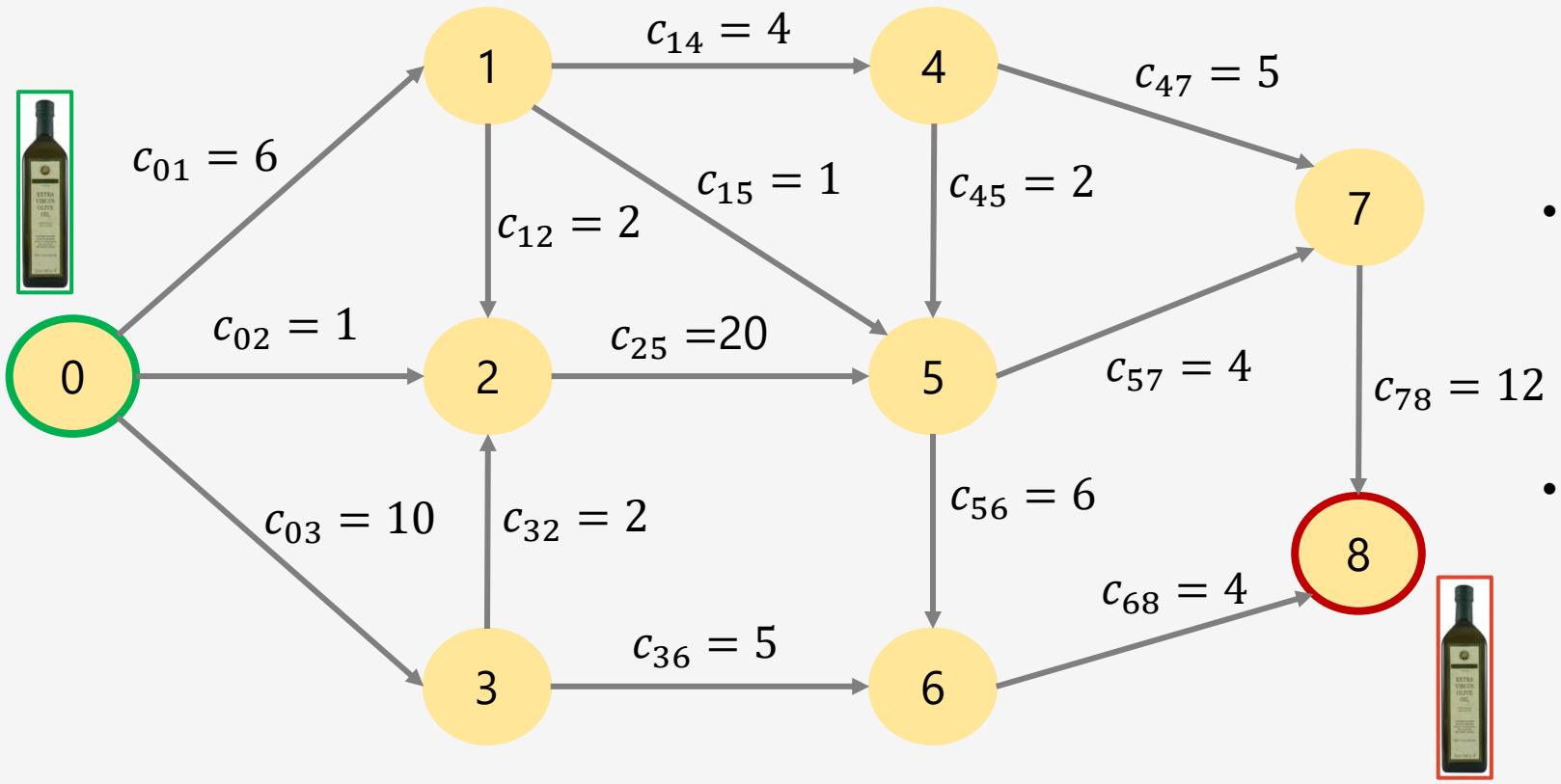
- **Objective**: transport all products produced in Plants $P_1, \dots, P_m$ to Warehouses $W_1, \dots, W_n$ with the <u>minimum cost</u>.

$$\min f(x) = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij}$$

- **Constraints**:
  - must satisfy the demand of each warehouse $\quad d_j(x) := \sum_{i=1}^{n} x_{ij} \geq D_j \qquad j = 1, \dots, m$
  - must not exceed supply capacity of each plant $\quad s_i(x) := \sum_{j=1}^{m} x_{ij} \leq S_i \qquad i = 1, \dots, n$

$$x_{ij} \geq 0, \forall i, j$$

# Examples of LPs: Maximum Flow Problem




- **Decision variables:** quantity of olive oil transported from node $i$ to node $j$.
- **Objective**: Ship as much olive oil as possible from the source node 0 to the terminal node 8.
- **Constraints:**
  - Cannot exceed the capacity of the edges
  - flowIN == flowOUT

Graph $G = (V, E)$ $\quad V = \{0,1,2,\dots,8\}$ $\quad E = \{(0,1),(0,2),\dots(7,8)\}$

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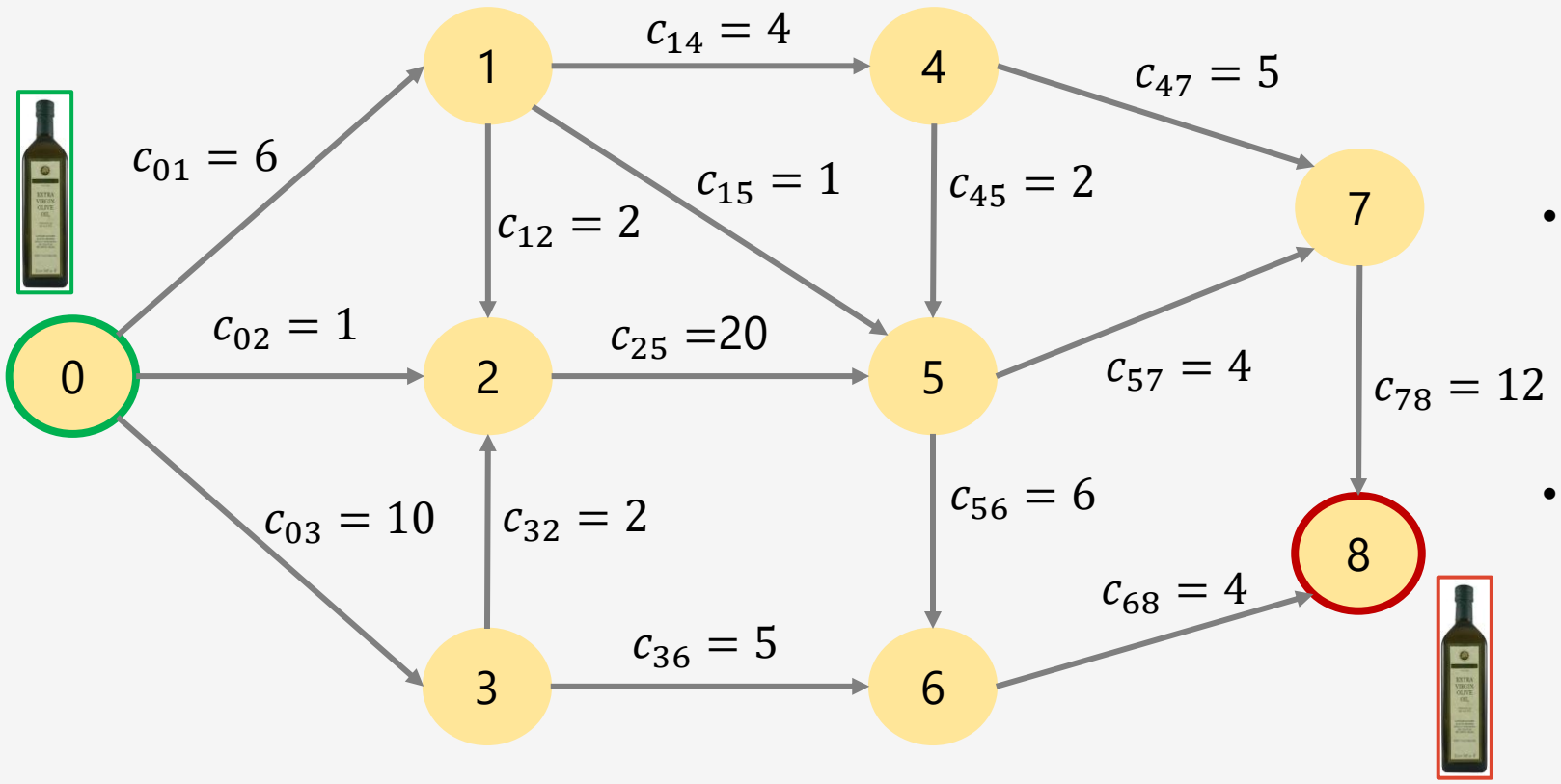


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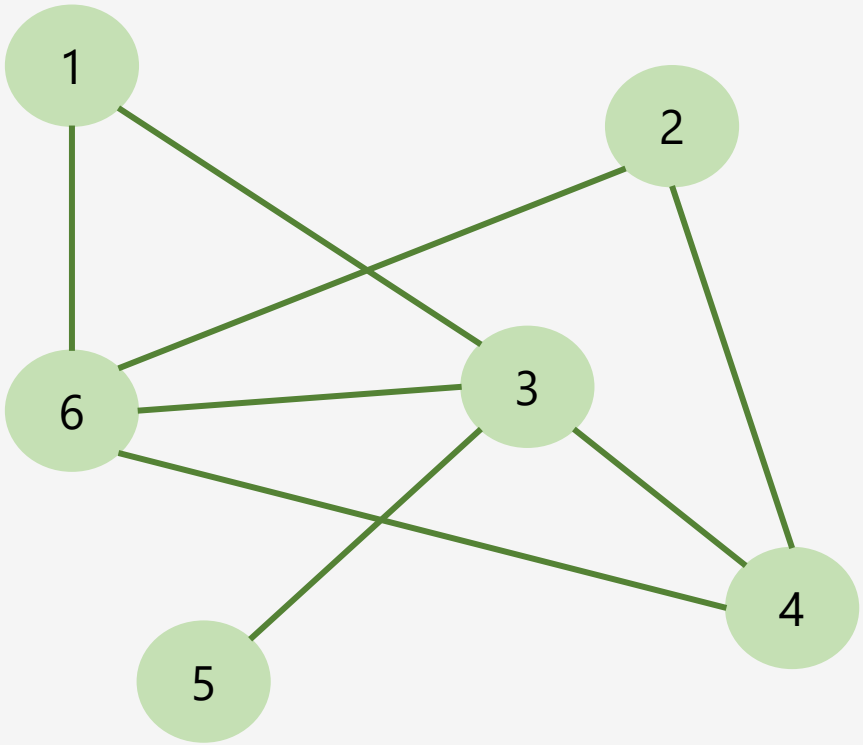
- **Decision variables:** quantity of olive oil transported from node $i$ to node $j$.
  - $x_{ij}$
- **Objective**: Ship as much olive oil as possible from the source node 0 to the terminal node 8.
  - $\max f(x) = x_{01} + x_{02} + x_{03}$
- **Constraints:**
  - Cannot exceed the capacity of the edges
    - $0 \leq x_{ij} \leq c_{ij}$
  - flowIN == flowOUT
    - $x_{01} = x_{14} + x_{15} + x_{12}$
    - $x_{14} = x_{47} + x_{45}$
    - $x_{15} + x_{45} + x_{25} = x_{57} + x_{56}$
    - …

# Examples of LPs: Largest Stable (or Independent) Set problem




- A **stable set** in an *undirected* graph $G = (V, E)$ is defined as a set of nodes $S \subset V$ with the property that the nodes of $S$ are **pairwise non-adjacent**
- Two nodes $i$ and $j$ are called **adjacent** iff $(i, j) \in E$
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- The **size of the largest stable set** of $G$ is denoted by $\boldsymbol{\alpha(G)}$
- We are looking for the **largest stable** set of $G$, that is the **largest** $\boldsymbol{\alpha(G)}$

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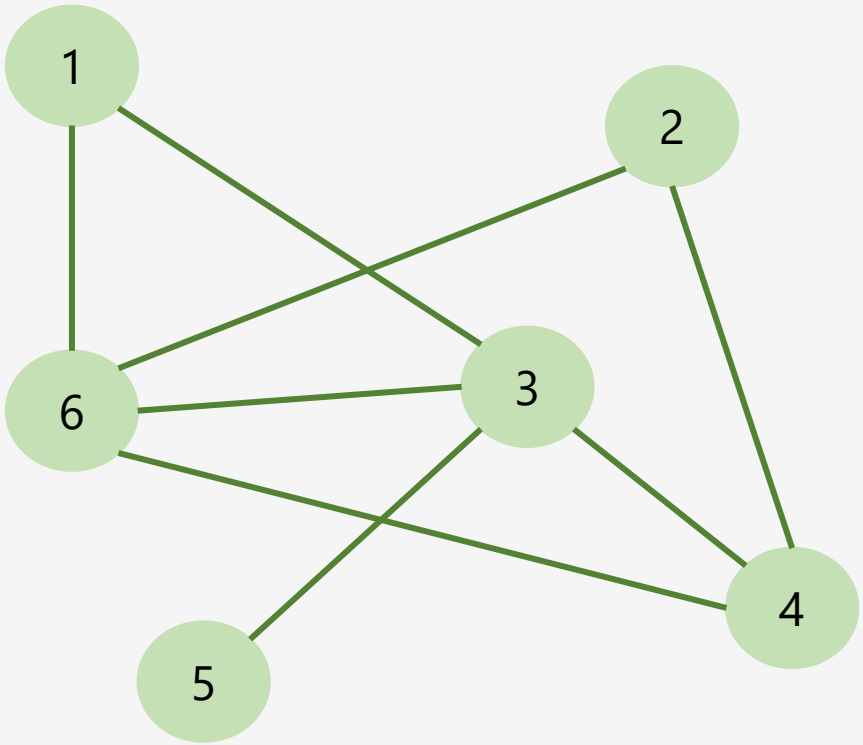


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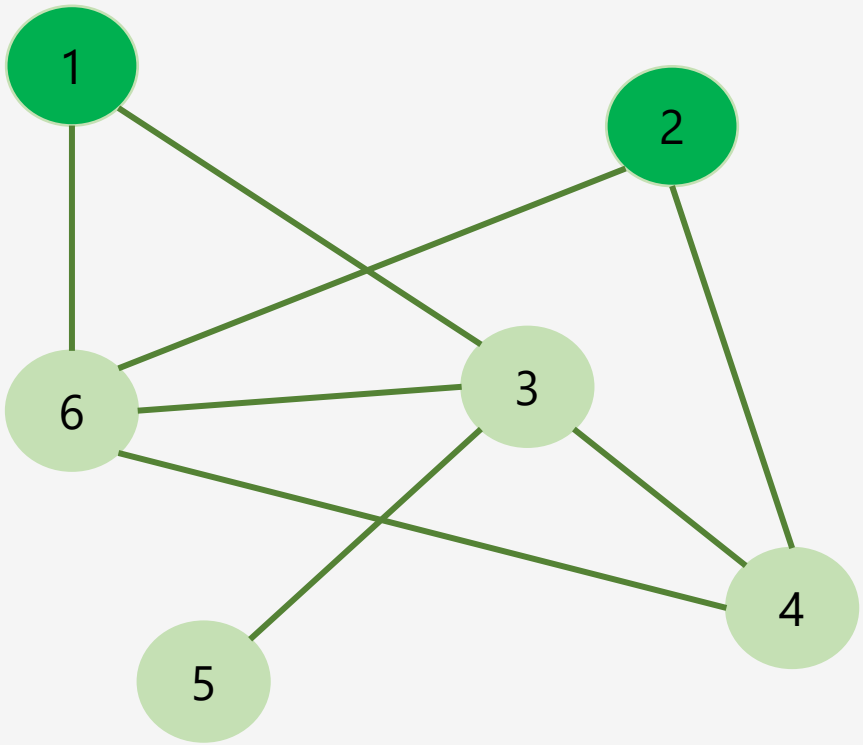


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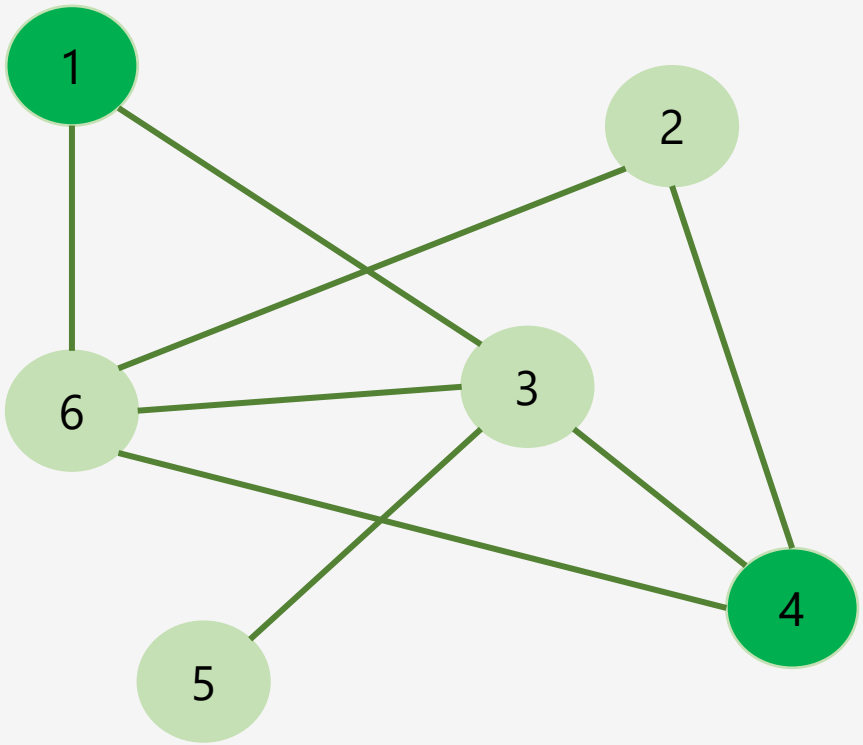


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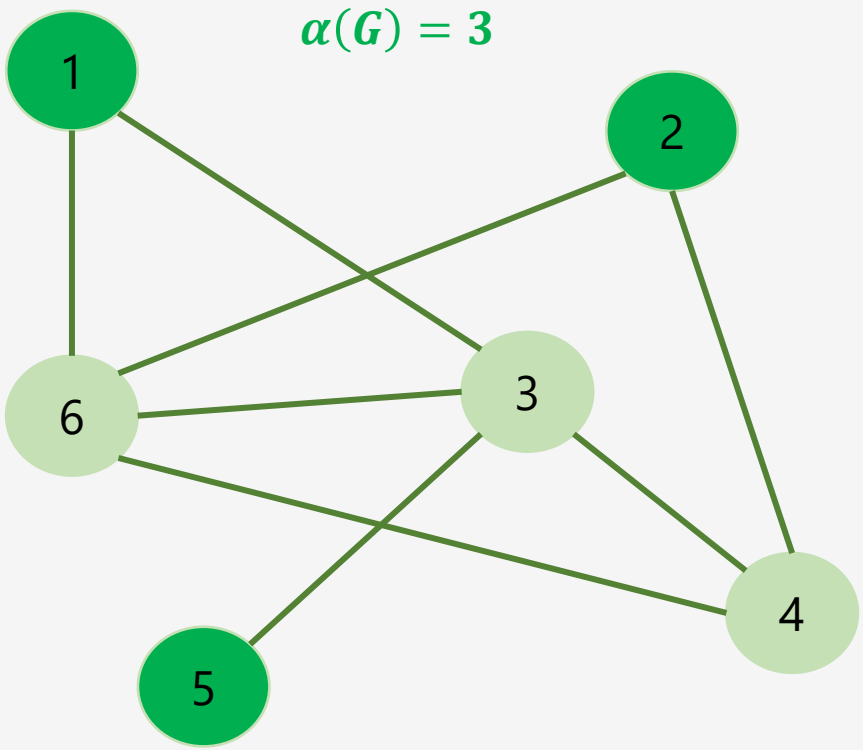


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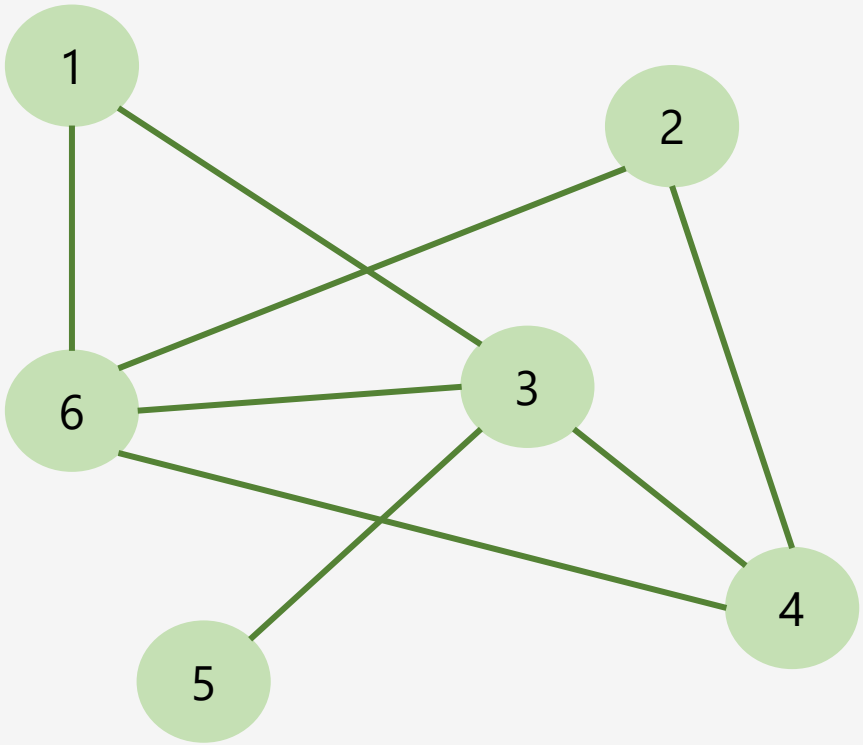


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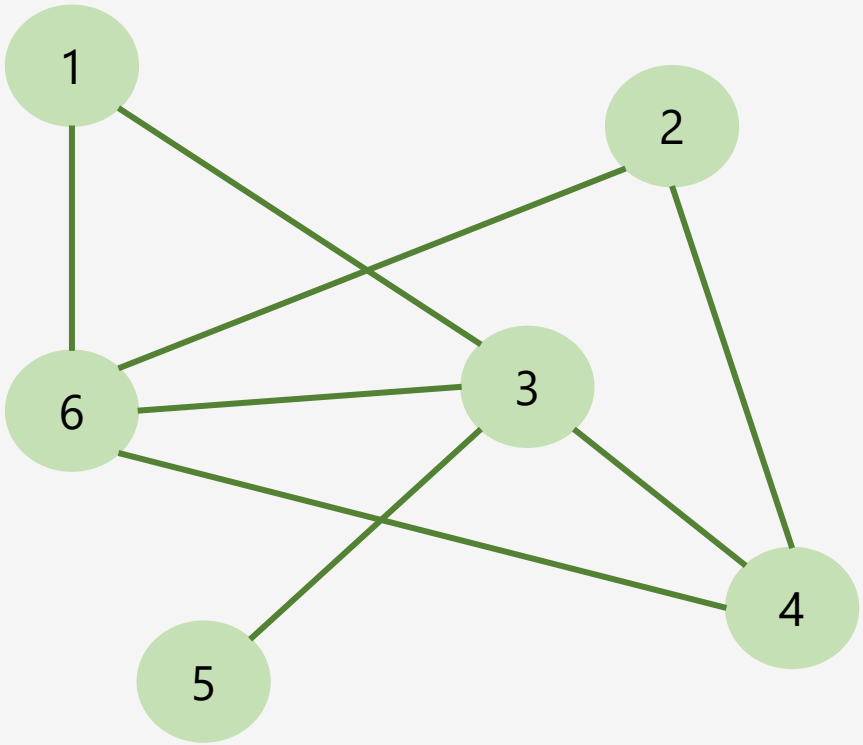
- **Decision variables:**
  - $x_i = 1$, if node $i$ is picked;
  - $x_i = 0$, otherwise
- **Objective:**
  - $\max f(x) = \sum_{i=1}^{6} x_i$
- **Constraints:**
  - $x_i + x_j \leq 1, \forall (i,j) \in E$
  - $x_i \in \{0,1\}, \forall i \in V$ → binary constraint

*Equivalent formulation of the binary constraints as continuous quadratic equations*

$$x_i(x_i - 1) = 0, \forall i \in V$$

# Examples of LPs: Largest Stable (Independent) Set problem




$$\left.\begin{array}{lll} \alpha(G) = max & & f(x) = \sum_{i=1}^{6} x_i \\ & s.t. & x_i + x_j \leq 1, \forall (i,j) \in E \\ & & x_i \in \{0,1\}, \forall i \in V \end{array}\right\} \quad (IP)$$

$$\left.\begin{array}{lll} LP_{opt} = max & & f(x) = \sum_{i=1}^{6} x_i \\ & s.t. & x_i + x_j \leq 1, \forall (i,j) \in E \\ & & 0 \leq x_i \leq 1, \forall i \in V \end{array}\right\} \quad (LP_{relax})$$

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$$\alpha(G) \leq LP_{opt}$$

**If $x^*_{LP}$ is the optimal solution of $LP_{relax}$ and $x^*_{LP} \in \{0,1\}^6 \Rightarrow \alpha(G) = LP_{opt}$**

# Examples of LPs

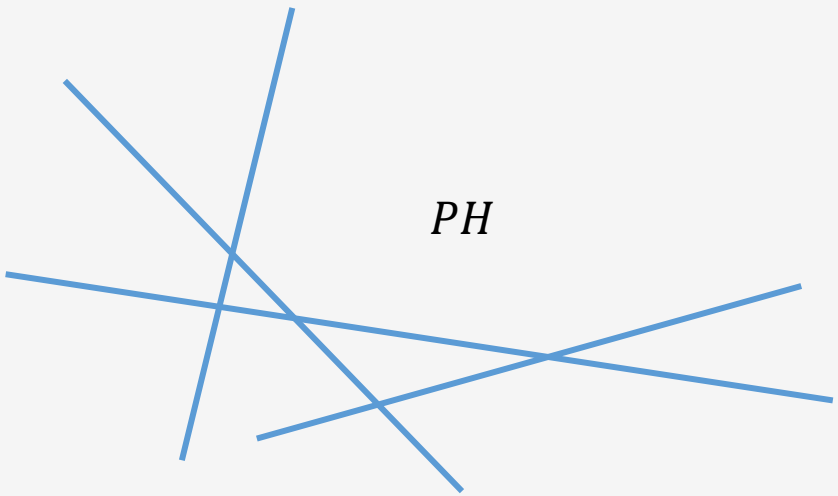
Study all LP examples in the notes (Lec11.pdf)

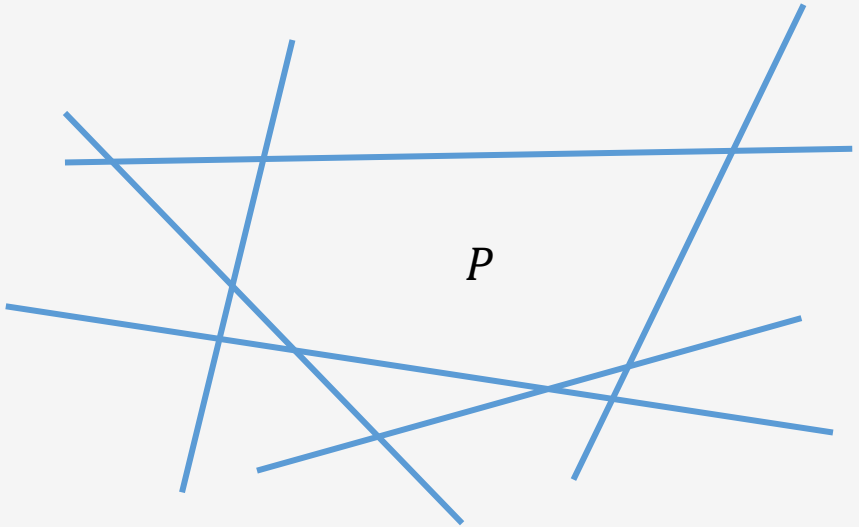
# What is a Polyhedron

- **Hyperplane**: $HP = \{x \in R^n \mid a^T x = b\}$
- **Halfspace**: $HS = \{x^{R^n} \mid a^T x \leq b\}$
- **Polyhedron**: Intersection of finitely many Halfspaces:

$$PH = \bigcap_{i=1}^{m} \{x \in R^n \mid a_i^T x \leq b_i\}, \quad \text{with } 0 < m < \infty$$

- **Polytope**: a <u>bounded</u> Polyhedron:
  - A set $P$ is <u>bounded</u> iff $\exists r > 0 : P \subseteq B(0, r)$
  - $B(0, r) = \{x \in R^n \mid ||x|| \leq r\}$ : Euclidean ball



# Geometry of LP

- LP has very rich geometrical structure.
- This structure has been exploited to develop computationally efficient algorithms for solving very large instances of LP
- We will present the fundamental geometrical results before we present the computational algorithm → Simplex

**Extreme point:** $\tilde{x} \in R^n$ is an extreme point of a **convex set** $S \subseteq R^n$ iff

- $\tilde{x} \in S$
- $\nexists y, z \in S \; with \; y \neq \tilde{x} \; and \; z \neq \tilde{x}$, and $\nexists \lambda \in [0,1]$ such that

$$\tilde{x} = \lambda y + (1 - \lambda) z$$

OR

- $\tilde{x} = \lambda y + (1 - \lambda) z$, for $y, z \in S$ and $\lambda \in [0,1] \Rightarrow \tilde{x} = y$ or $\tilde{x} = z$

It is used to prove the Optimality Theorem for LPs

**Vertex:** $\hat{x} \in R^n$ is a vertex of the **Polyhedron**

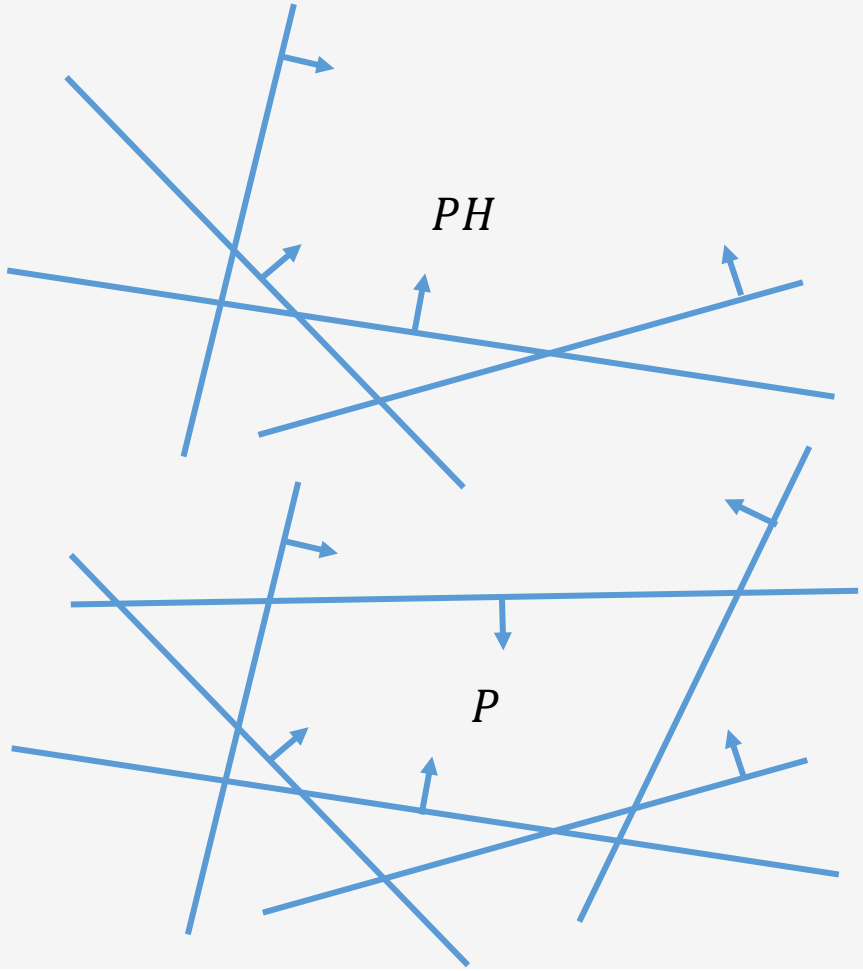
$$P = \{x \in \mathrm{R}^n | \; a_i^T x \leq b_i, i = 1, \dots, m\} \text{ iff}$$

- $\hat{x} \in P$
- $\exists \; n$ linearly independent constraints that are tight at $\hat{x}$
  - A constraint $a^T x \leq b$ is **tight** at $\hat{x}$ IFF $a^T \hat{x} = b$

It is used to construct an algorithm (Simplex) for efficiently solving LPs

# Geometry of LP

What are the **extreme points** and **vertices** of $PH$ and $P$ ??



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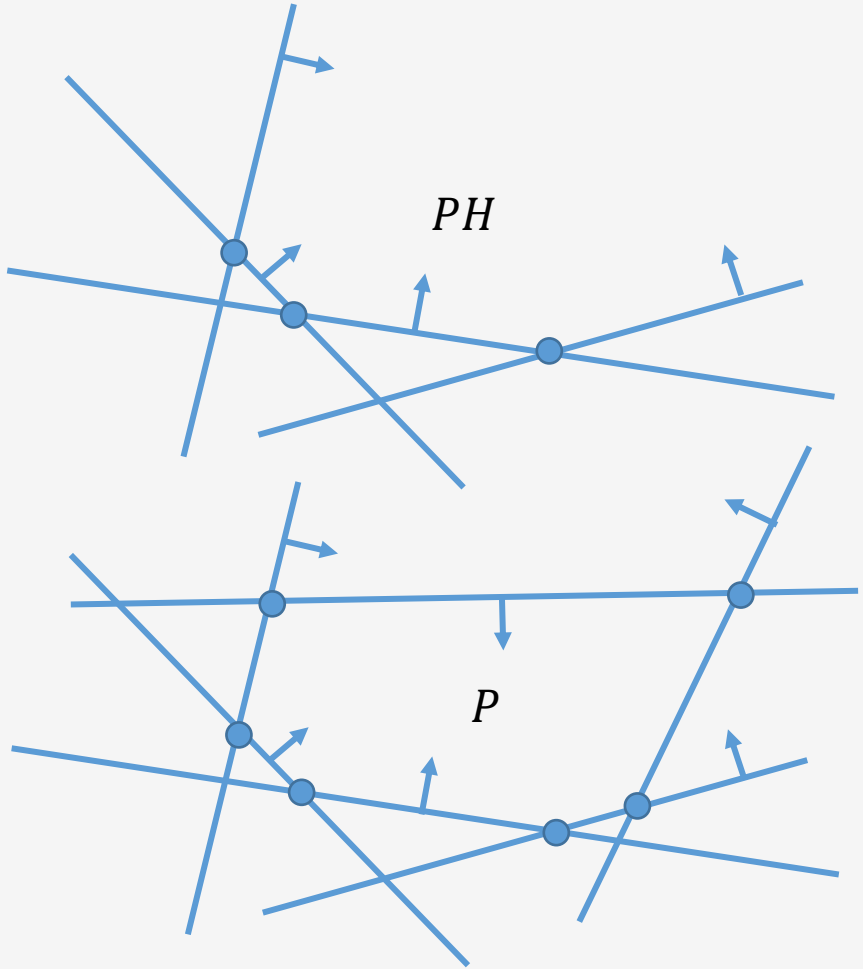

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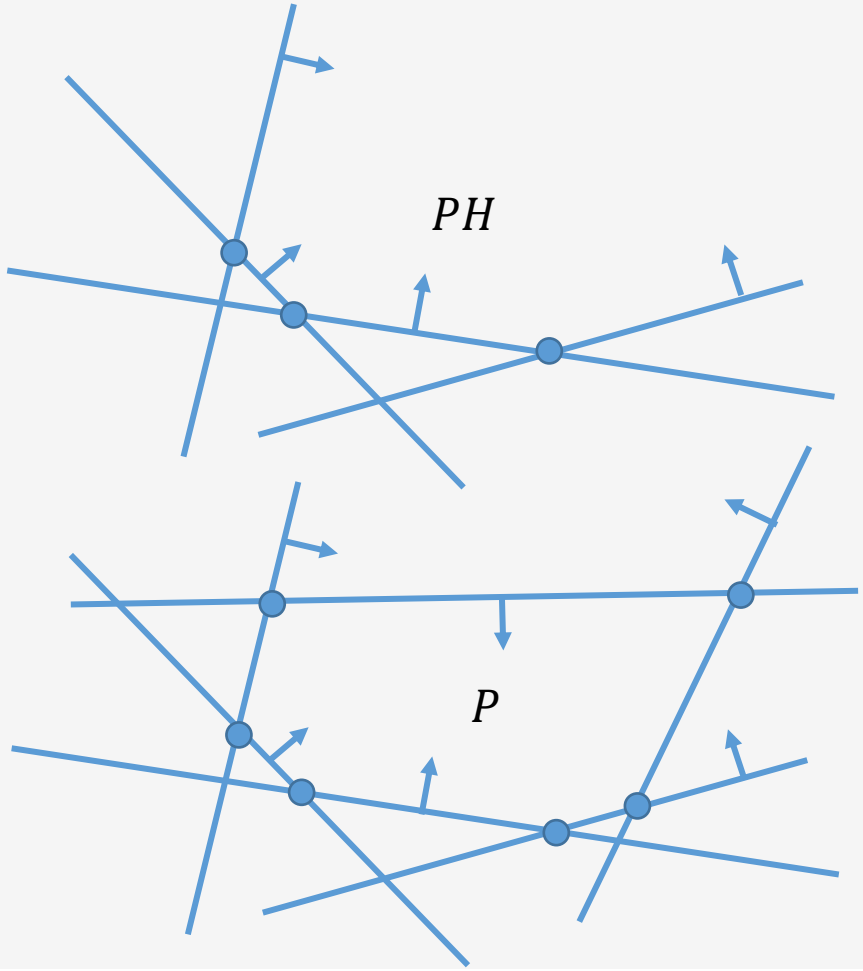


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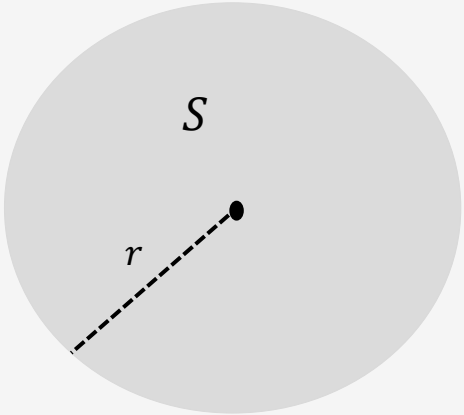

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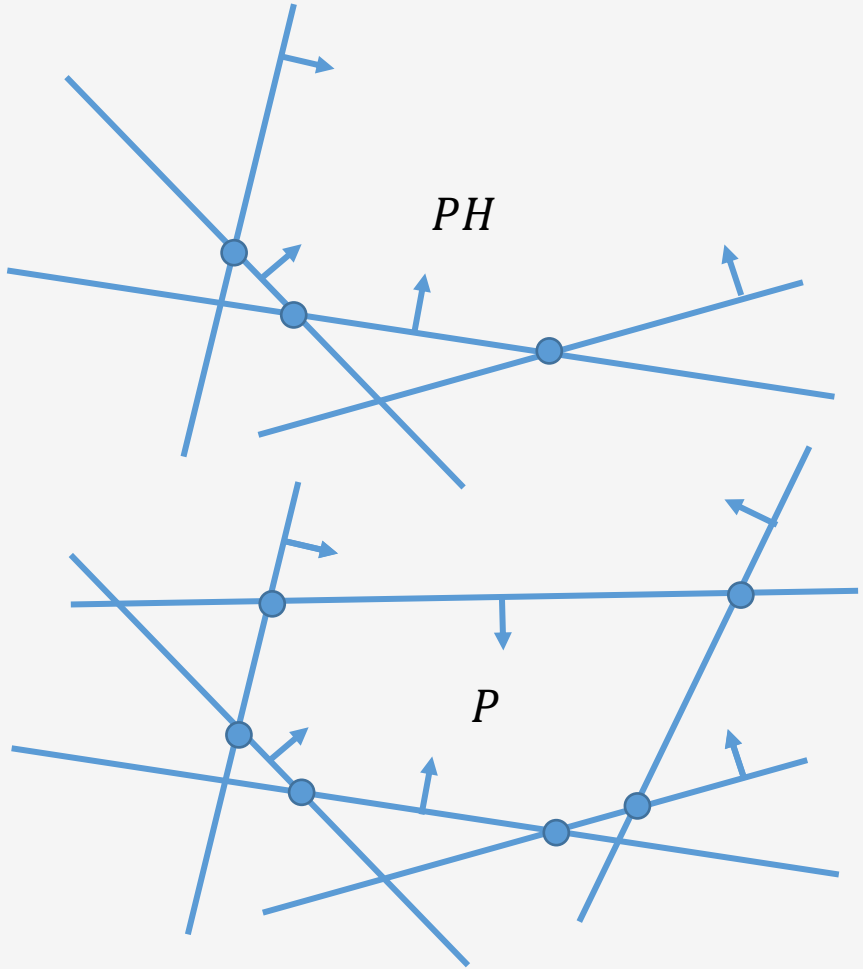


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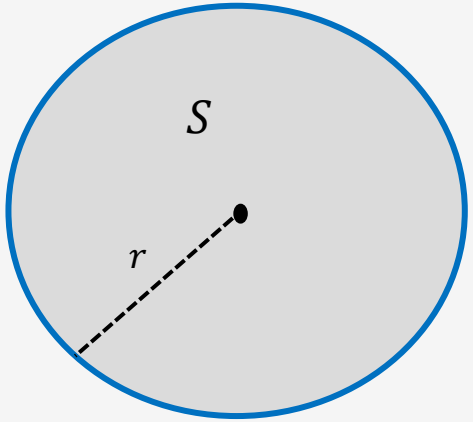
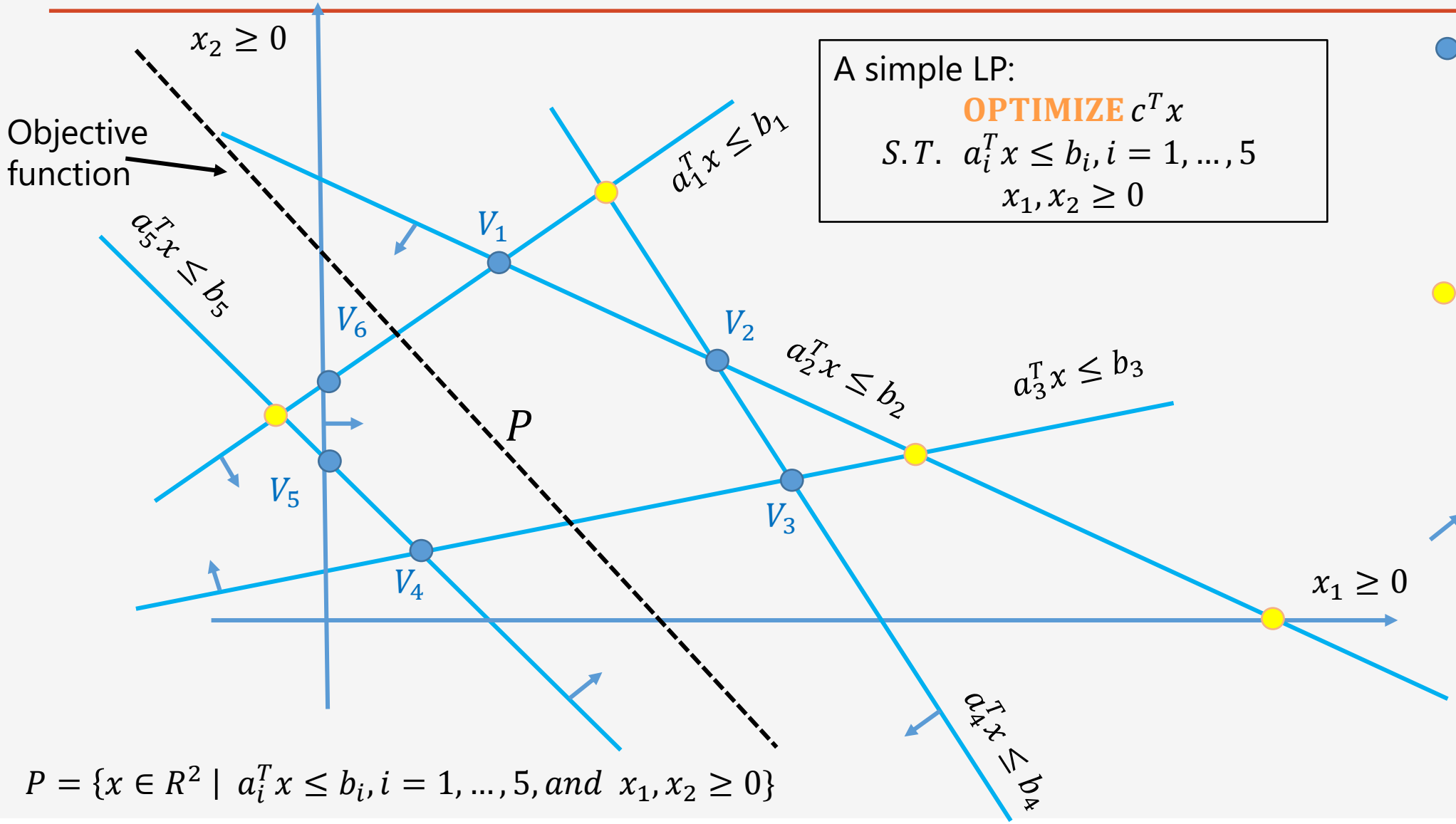



All points on the **boundary** are extreme points.
There are **no vertices**, **WHY??**

# Visualizing the solution process on a small LP




A simple LP:

$$\text{OPTIMIZE } c^T x$$
$$S.T.\ \ a_i^T x \leq b_i, i = 1, \dots, 5$$
$$x_1, x_2 \geq 0$$

Vertex, because it satisfies both conditions

Not a Vertex, because it is not feasible point

Which half-space we consider

$$P = \{x \in R^2 \mid a_i^T x \leq b_i, i = 1, \dots, 5, and\ \ x_1, x_2 \geq 0\}$$

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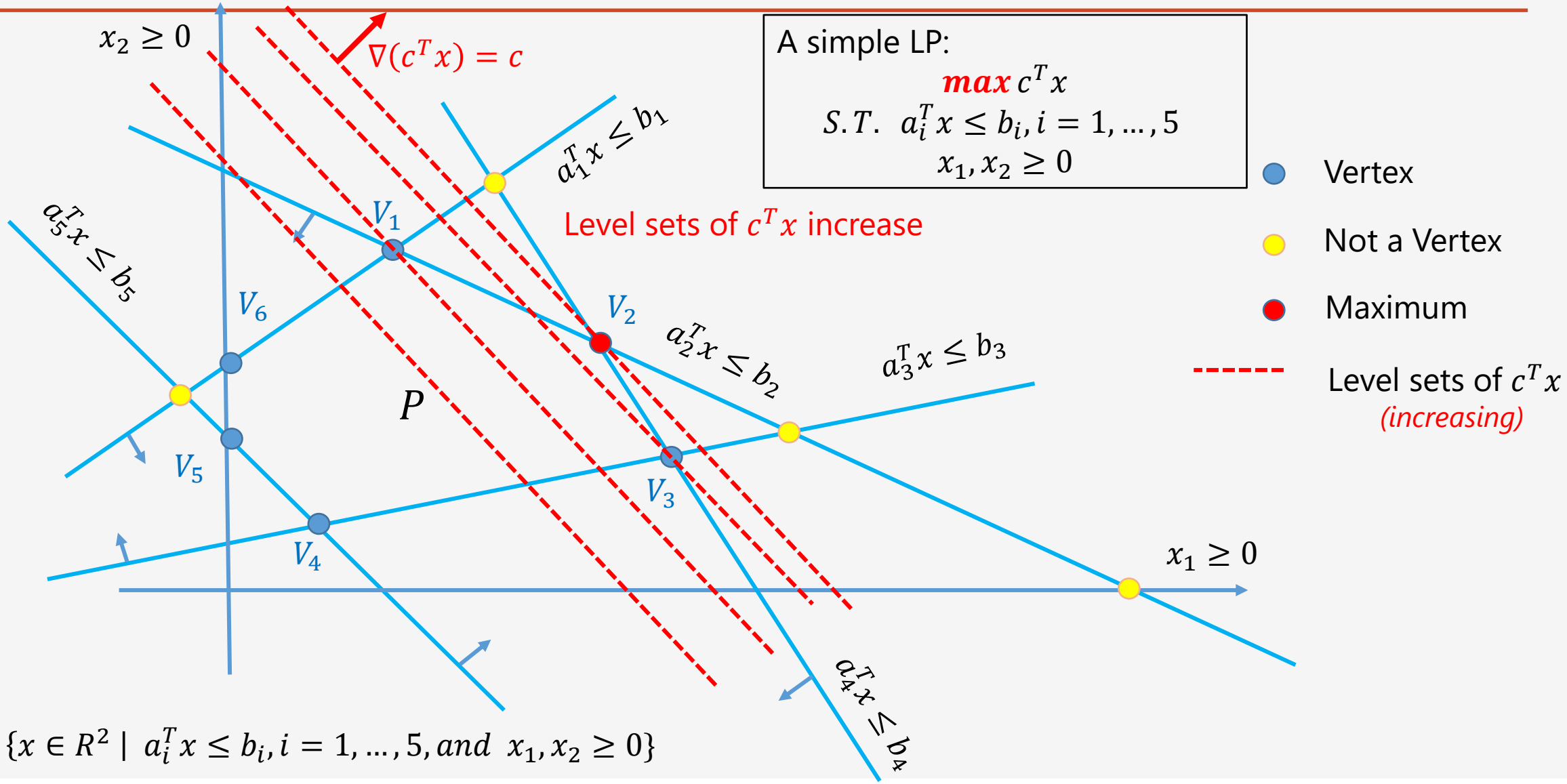


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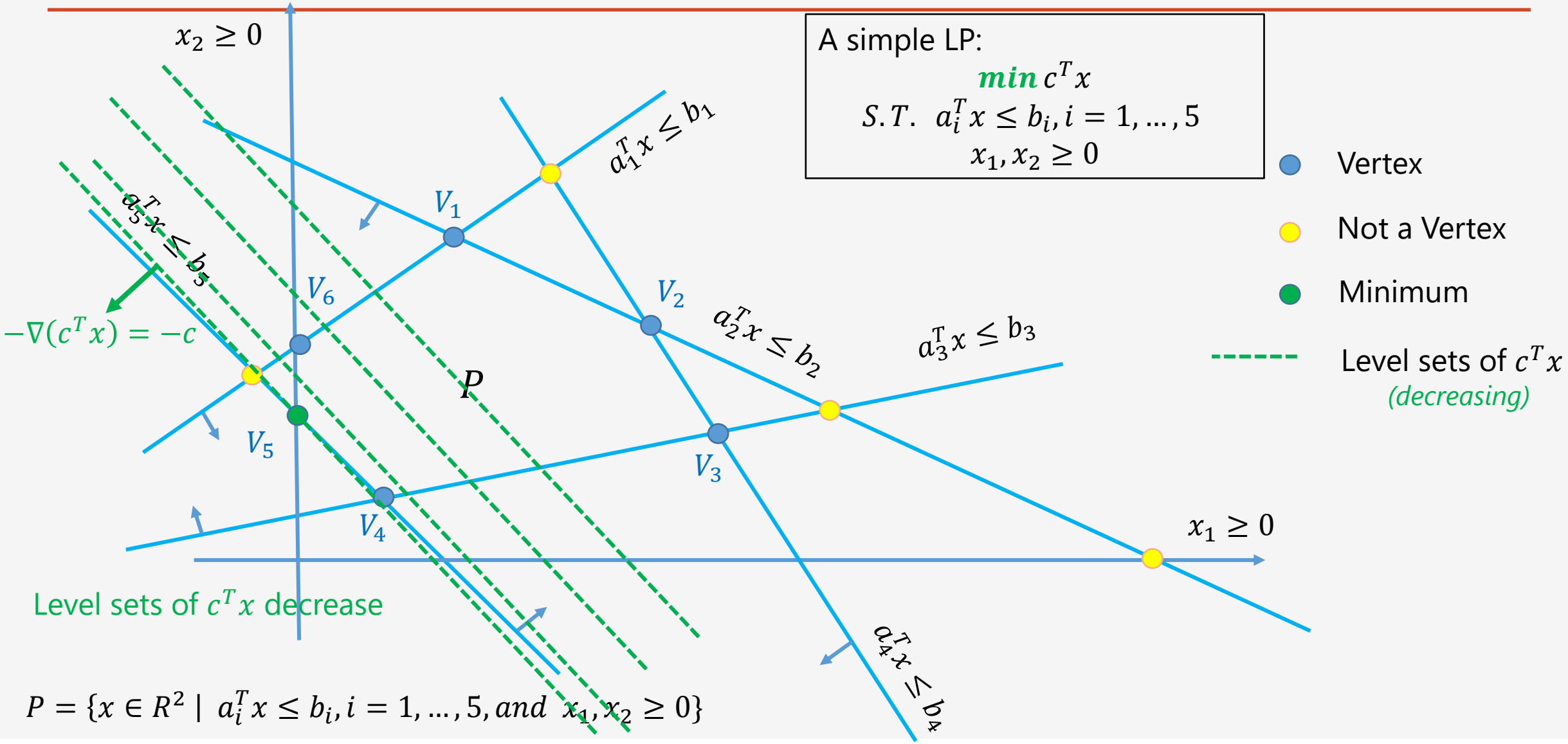


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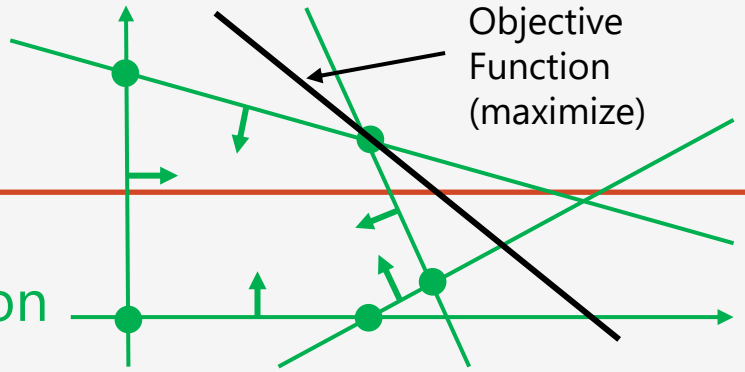
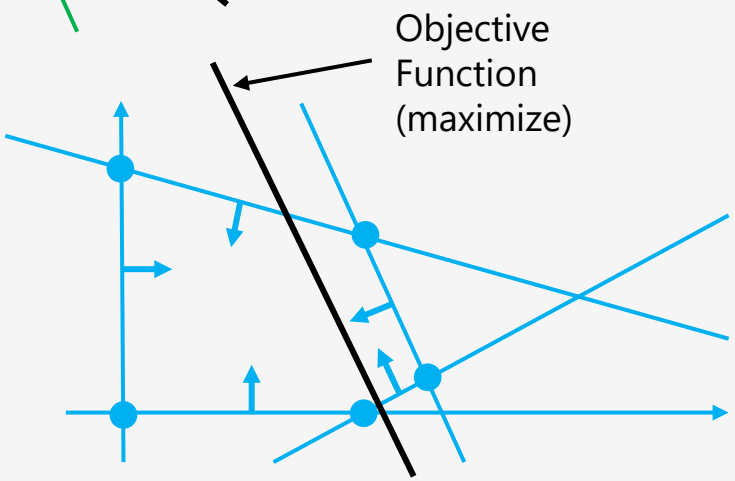
# There are 4 Types of LPs

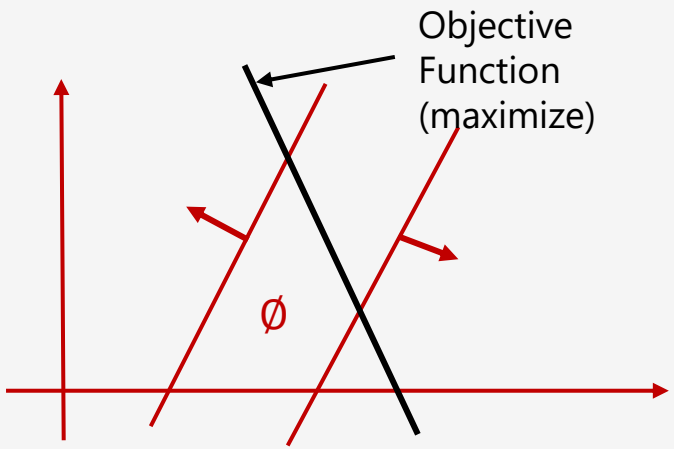
1. LPs with unique optimal solution



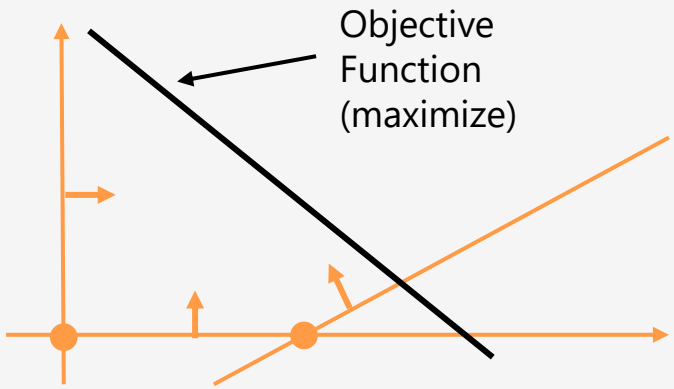

2. LPs with ∞ number of solutions




3. Infeasible LPs




4. Unbounded LPs



# Extreme points & Vertices in a Polyhedron

**Theorem:** Let $P = \{x \in R^n \mid Ax \leq b\}$ be a non-empty polyhedron, where $A \in R^{m \times n}$, and $\hat{x} \in P$.

Then we have: $\hat{x}$ is an extreme point of $P$ $\Leftrightarrow$ $\hat{x}$ is a vertex pf $P$

**Proof:** See pdf notes

**Remarks:** Simple but important result because:

1. We use this theorem to prove optimality for LPs

2. We use the vertex to produce an algorithm that converges to the optimal solution of an LP

**Question:** Does every Polyhedron $P$ have an extreme point??

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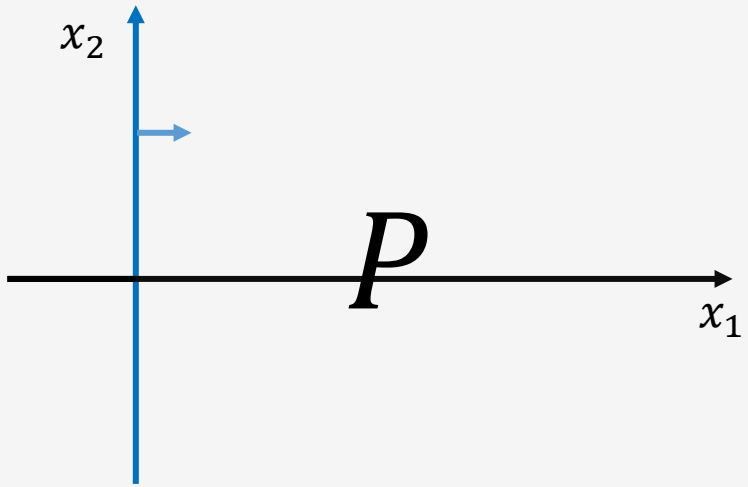
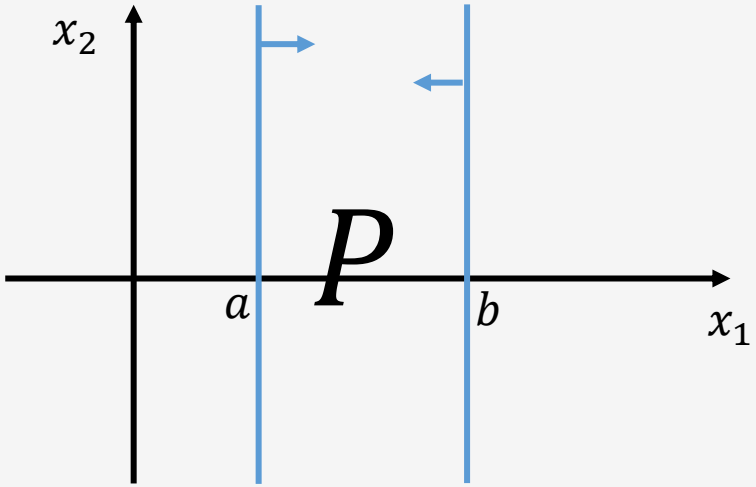



$P = \{x \in R^2 | x_1 \geq 0\}$

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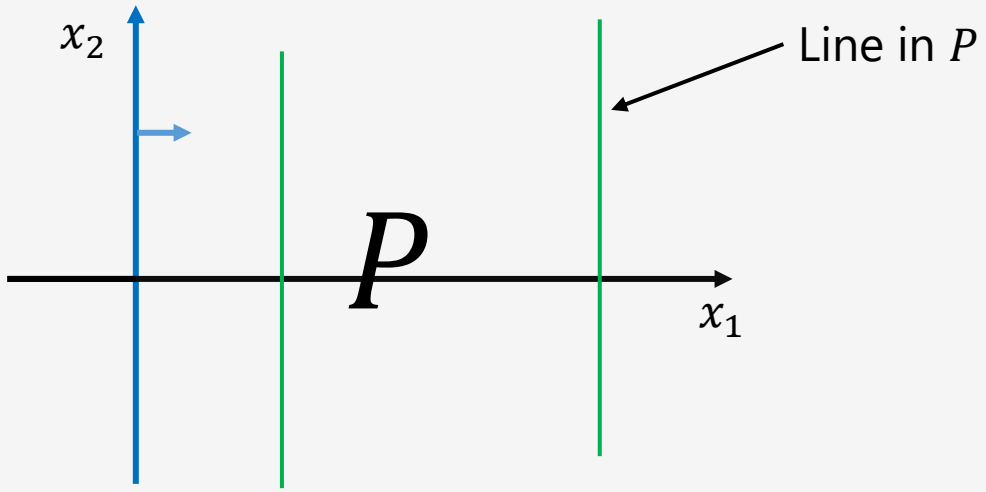


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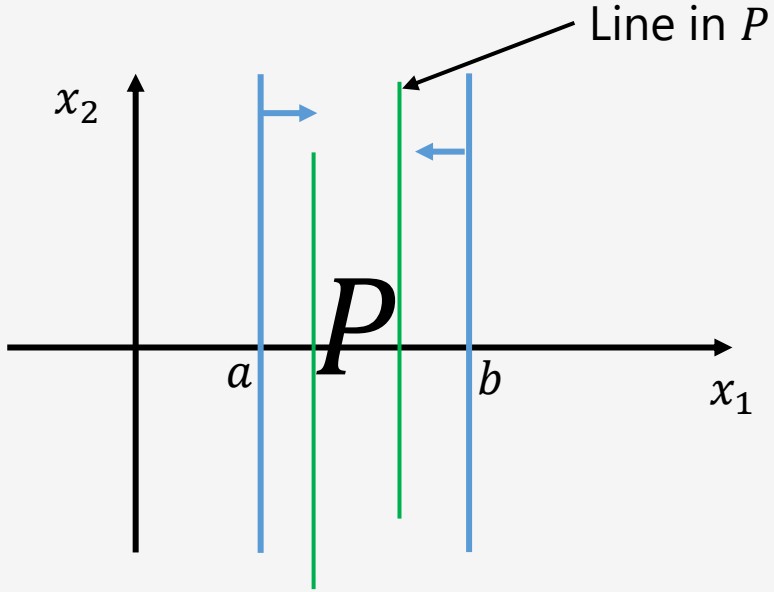


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# Extreme points & Vertices in a Polyhedron

**Theorem**: A polyhedron $P$ has at least one extreme point **IFF** it does not contain a line

**Question**: Can you think of a polyhedron that cannot contain a line??

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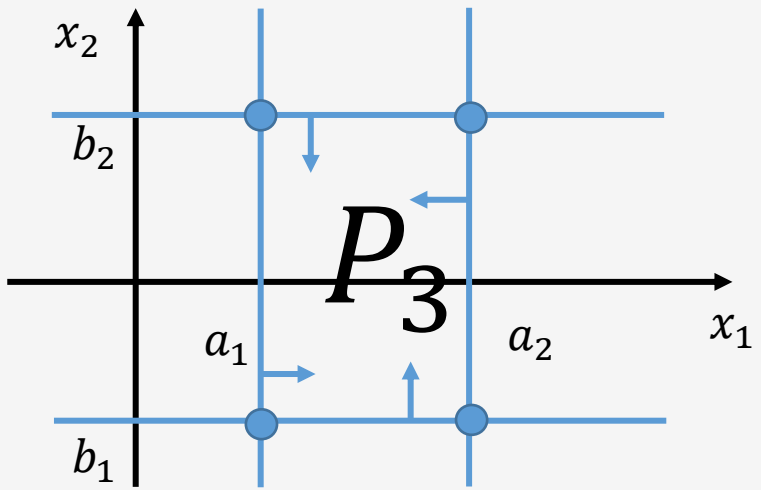


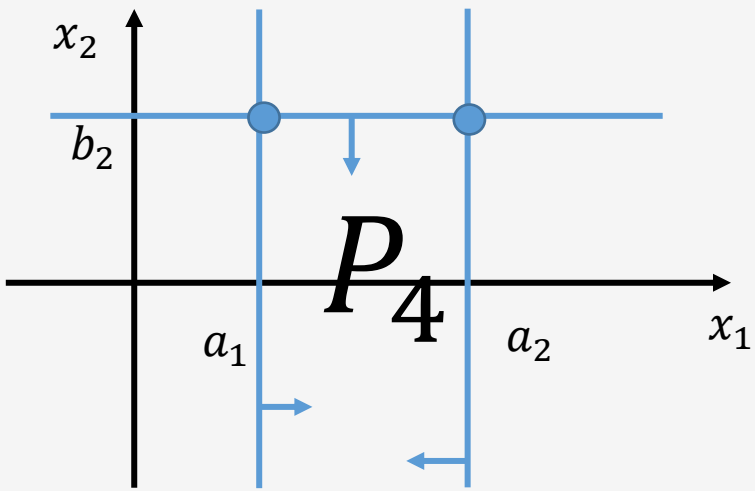


$$P_3 = \{x \in R^2 |\ \ a_1 \leq x_1 \leq a_2,\ \ b_1 \leq x_2 \leq b_2\}$$

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# Extreme points & Vertices in a Polyhedron

**Corollary:** Every Polytope has an extreme point

**Remark:** How can we represent the phrase: "it contains a line" in *mathematical form?*

$$\exists \boldsymbol{x} \in \boldsymbol{P}, \boldsymbol{d} \in \boldsymbol{R}^{\boldsymbol{n}} \textbf{ such that}: \quad \boldsymbol{x} + \boldsymbol{\alpha d} \in \boldsymbol{P}, \forall \boldsymbol{a} \in \boldsymbol{R}$$

# Optimality and Extreme Points in LP

**<u>Theorem</u>** (optimality @ extreme points): Consider the LP:

$$[\ \min f(x) = c^T x \quad s.t. \quad Ax \leq b\ ]$$

where $A \in \mathrm{R}^{m \times n}$ and $b \in R^m$. Denote the feasible region of the LP as

$$P = \{x \in R^n \mid Ax \leq b\}.$$

*<u>Assume</u>* that $P$ has at least one extreme point.

*IF there exists an <u>optimal solution</u>, THEN it is an <u>extreme point</u> of $P$*

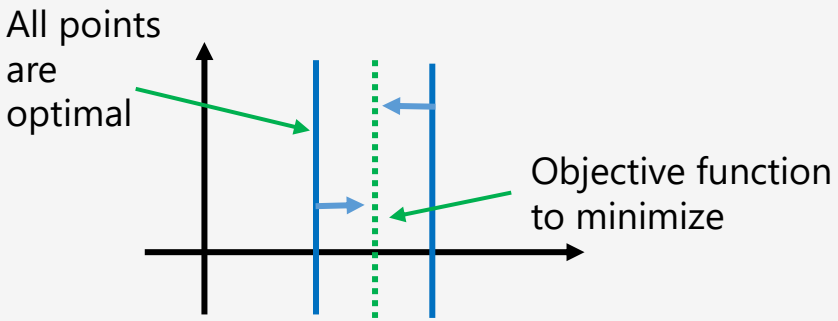
# Optimality and Extreme Points in LP

**Discussion before the proof**

1. **Why do we need to assume that $P$ has at least one extreme point??**

   a) To exclude the case where we have many optimal solutions,

   *but none is extreme*



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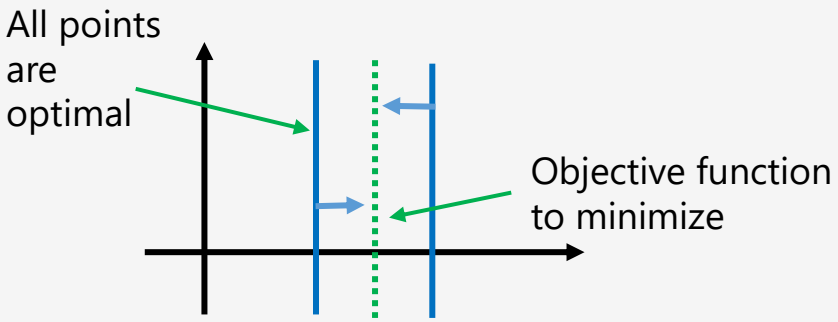


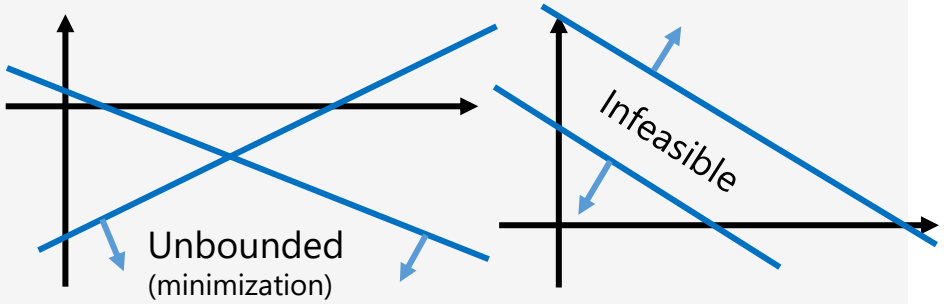

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   a) To exclude the case where we have many optimal solutions,

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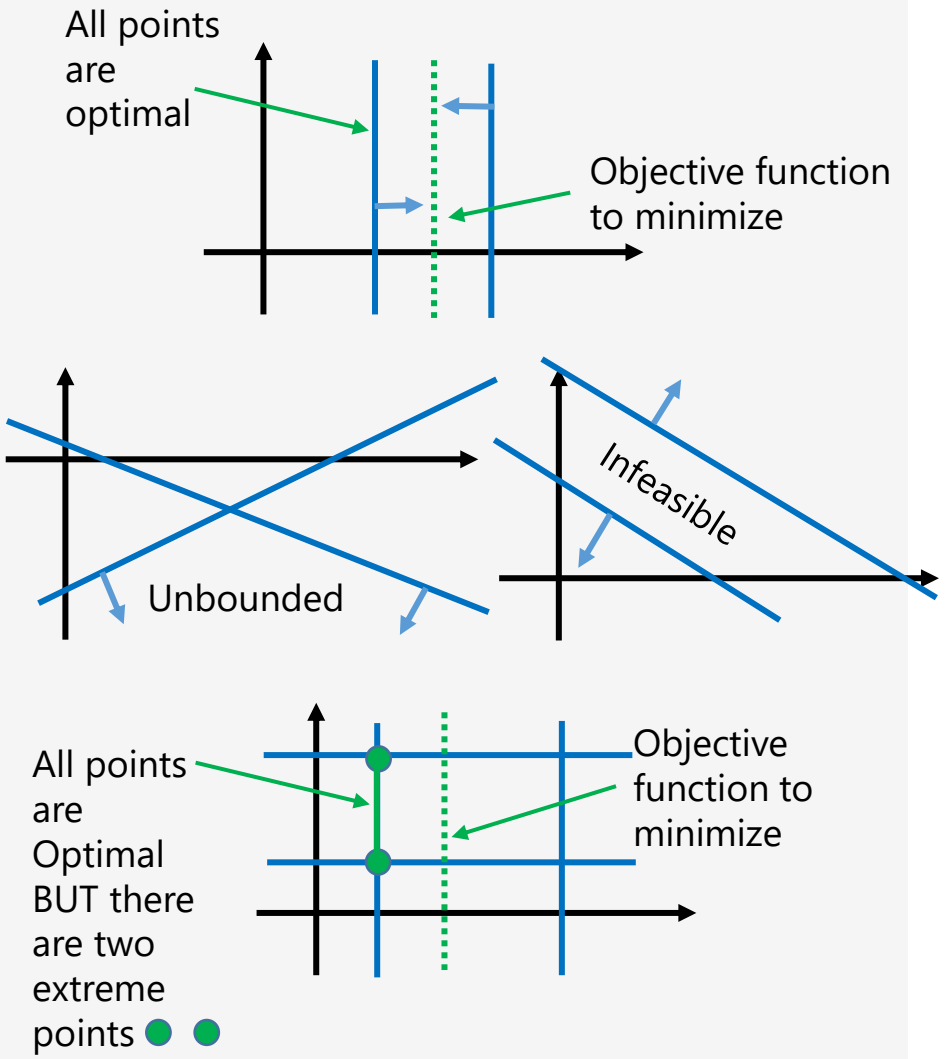
2. **Why do we need to assume that there exists an optimal solution??**

   a) To exclude that $P$ is neither unbounded nor infeasible

3. **Important! We are not saying that all optimal solutions are extreme!**

   a) We may have an infinite number of optimal solutions,

   but at least one is extreme point



# Optimality and Extreme Points in LP

**Proof of the Theorem:** Recall that the feasible region of the LP is $P = \{x \in R^n \mid Ax \leq b\}$

Let $Q \subseteq R^n$ be the **set of all optimal solutions** to the LP $\Rightarrow$ $\boldsymbol{Q \subseteq P}$ (→ optimal solutions are feasible)

If $v \in R$ is the **optimal value** of the LP then $Q$ is defined as: $Q = \{x \in R^n \mid Ax \leq b \;\; and \;\; c^T x = v\}$

Because of our assumption: the set $P$ has an extreme point $\Rightarrow$ $P$ does not contain a line $\Rightarrow$

$\Rightarrow$ $Q$ does not contain a line $\Rightarrow$ $Q$ has an extreme point

Now let $x^*$ be an extreme point of $Q$ → $x^*$ is optimal (minimum) solution → $v = c^T x^*$

We will prove that $x^*$ is an extreme point of $P$, as well (we will do that by *contradiction*)

# Optimality and Extreme Points in LP

**Proof of the Theorem: (Cont.)** **Assume that $x^*$ is not an extreme point in $P$** $\Rightarrow$

$$\exists y, z \in P, \quad y \neq x^*, \quad z \neq x^*, \; and \;\; \exists \lambda \in (0,1) \quad \text{such that} \quad x^* = \lambda y + (1-\lambda) z$$

Multiply both sides by $c^T$

$$v = c^T x^* = \lambda c^T y + (1-\lambda) c^T z$$

This is strange!! 😐 Because we know that $c^T x^* = v \leq c^T y$ (with $y \neq x^*$) and $c^T x^* = v \leq c^T z$ $(with\ z \neq x^*)$, so the convex combination of $c^T y$ and $c^T z$ **cannot** be equal to $v$ (contradiction!) ■

**Example:** $5 = \lambda 5.1 + (1-\lambda) 5.2 = \lambda 5.1 - \lambda 5.2 + 5.2 = -\lambda 0.1 + 5.2, \quad \forall \lambda [0,1)$ (impossible!)

# Optimality, Extreme Points and Algorithmic Solutions

**How can I use the Theorem to find the optimal solution of an LP ??**

# Optimality, Extreme Points and Algorithmic Solutions

**How can I use the Theorem to find the optimal solution of an LP ??**

I could evaluate the objective function on all the vertices of the feasible region of the LP and select the one that gives me the best objective value

# Optimality, Extreme Points and Algorithmic Solutions

**A general simplistic algorithm:**

1. Choose $n$ out of the $m$ constraints $\binom{m}{n}$ combinations
2. Convert them to equalities (by adding slack variables)
3. Call a linear equations solver we already know (e.g., Gaussian elimination or a QP solver)
4. The linear solver will give me a solution, say $\hat{x} \in R^n$
5. Check if $\hat{x}$ satisfies the remaining $(m-n)$ inequalities
   a) If YES, Then $\hat{x}$ is a vertex → store $\hat{x}$ in the "bag of vertices" $B$ $(\hat{x} \rightarrow B)$
   b) If NO, Then $\hat{x}$ is not a vertex (it is infeasible point) → ignore it.

- **Repeat** the above steps

# Optimality, Extreme Points and Algorithmic Solutions

- At the end of this process the vertex $x^* \in B$ that gives me the highest value of the objective (if we maximize) will be the optimal solution $(c^T x^* \geq c^T x, \forall x \in B)$
- This is a very simple algorithm, and it will always converge to the optimal solution
- **But what is its weak point???**

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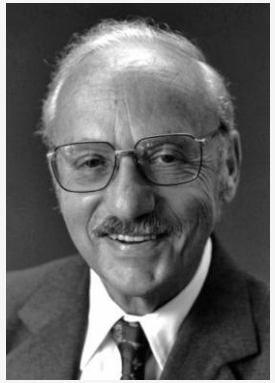
- This is a very simple algorithm, and it will always converge to the optimal solution
- **But what is its weak point???**
- It can be very slow as in general it will need to visit *all vertices* of the feasible region (not polynomial time algorithm)
- The **Simplex algorithm** tries to generate ***iterations that visit as few vertices as possible***
  - But in general, there is **no proof** that shows that the Simplex algorithm is **polynomial time** …

# The Simplex algorithm for LP

It was developed by **George Dantzig**, in the 1940s



He thought it was a **homework assignment** by his Statistics professor

Some of the movie scenes in **"Good Will Hunting"** is based on Dantzig's story

At each iteration, the Simplex algorithm performs **2 main tasks:**

1. **Check** if the current vertex is **optimal**?

2. **IF** yes, **THEN** Stop (you found the optimal solution); **ELSE** determine which **neighboring vertex** to move to next

**Definition:** Two vertices are **neighbors** if they share $\boldsymbol{n-1}$ **tight constraints**

**Observation #1:** Both tasks are easier to do if the current vertex is the Origin ($V_k \rightarrow x_k = 0$)

**Observation #2:** If the vertex is not the origin, we can change coordinates and transform it to the origin.

# The Simplex algorithm for LP

Recall the LP in standard form:

$$\begin{array}{ll} max & c^T x \\ s.t. & Ax = b \\ & x \geq 0 \end{array}$$

**Theorem:** The origin $x = 0$ is optimal (maximum) $\Leftrightarrow$ $c_i \leq 0,\ for\ all\ i = 1,2,\dots,n$

**Proof:**

$(\Leftarrow)$

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($\Leftarrow$) We have $[c_i \leq 0,\ \forall i\ \ and\ \ x_i \geq 0\ \forall i]\ \Rightarrow\ c^T x \leq 0\ \Rightarrow\ \ x = 0$ is optimal because **0 is the largest value** we can achieve (recall that we have a maximization problem)

($\Rightarrow$) Next slide ……

# The Simplex algorithm for LP

**Proof (cont.)** (⇒) We have that: $x = 0$ is optimal and want to show $c_i = 0, \ \forall i$

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Assume: $\exists j \in \{1,2,\dots,n\}$ such that $c_j > 0$

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As a result, the point $\tilde{x} = \left(0, \dots, 0, \epsilon_j, 0, \dots 0\right)^T$ will have a larger objective value ($c^T \tilde{x} = c_j \epsilon_j > 0$) than that of $x = 0$ ($c^T x = 0$)

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But this will violate the fact that $x = 0$ is optimal → Contradiction! Hence $c_i \leq 0, \forall i$ ■

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**Remark:** This is the reason we claimed that both tasks of the Simplex algo are easy to check if the current vertex is the origin ($x = 0$).

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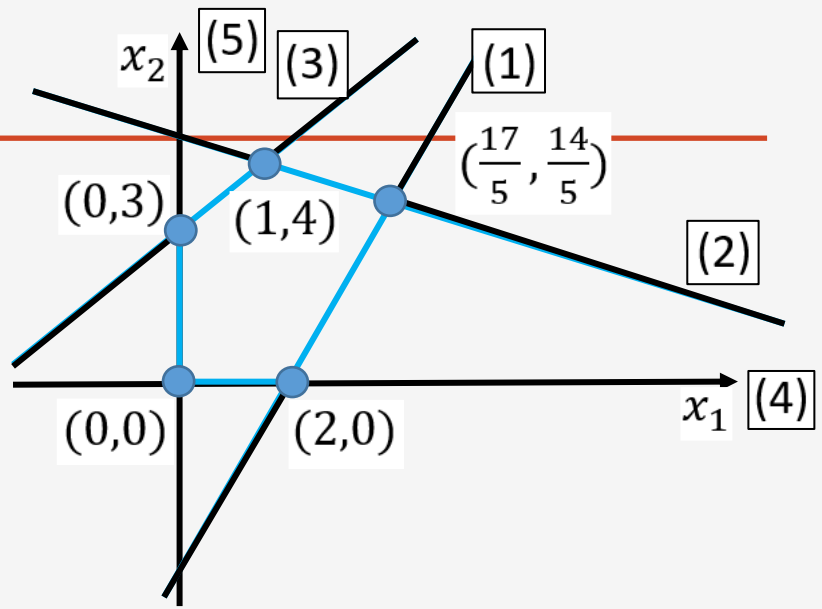


**Example:**

$$\max_{x \in R^2} \; 2x_1 + 5x_2,$$

$$\begin{aligned} s.t. \quad & 2x_1 - x_2 \leq 4 && (1) \\ & x_1 + 2x_2 \leq 9 && (2) \\ & -x_1 + x_2 \leq 3 && (3) \\ & x_1 \geq 0\,, && (4) \\ & x_2 \geq 0. && (5) \end{aligned}$$

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- We start at the origin $(x_1, x_2) = (0,0)$ which is a vertex since it makes constraints (4) and (5) tight.
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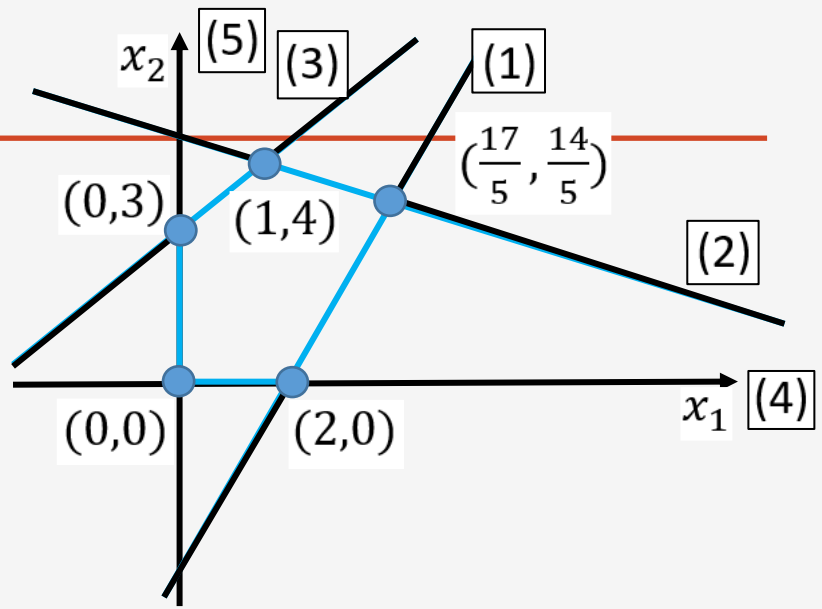


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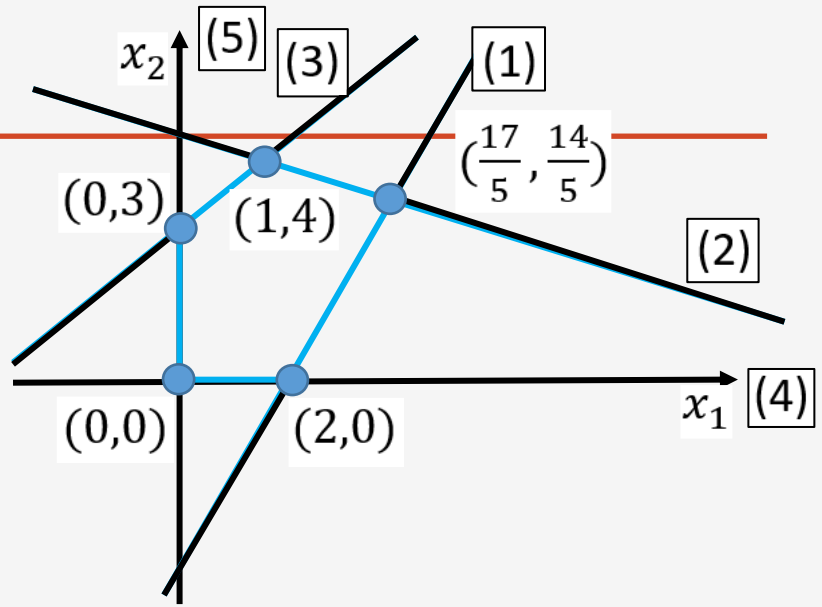


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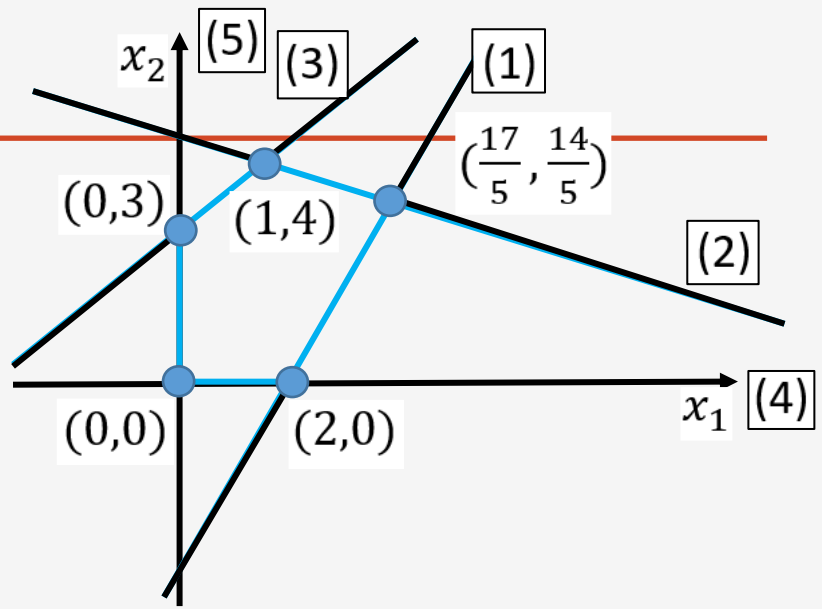


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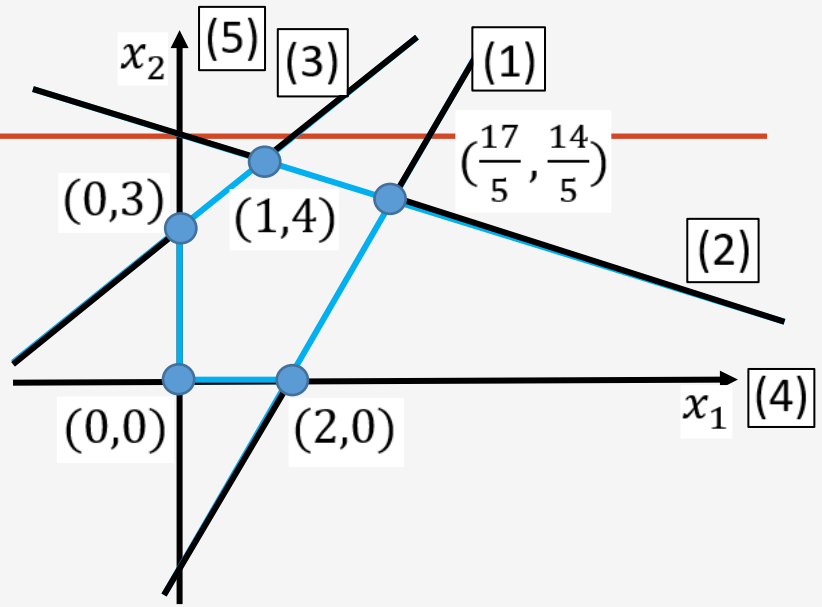
- We do that by releasing constraint (5) → this means that we can increase variable $x_2$, while keeping $x_1 = 0$ (it still remains tight)
- By increasing $x_2$ we will meet first constr. (3) and then constr. (2) → We must stop when we meet constr. (3) and hence $x_2 = 3$

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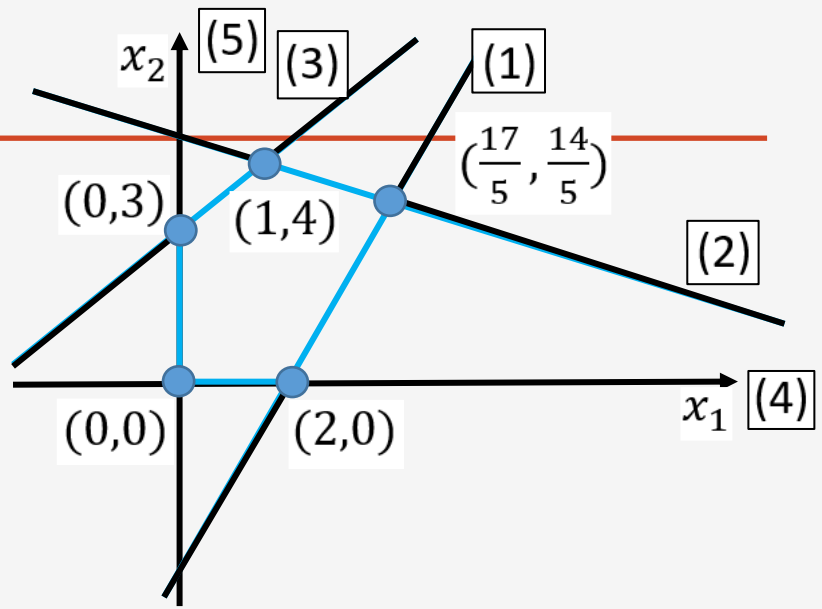


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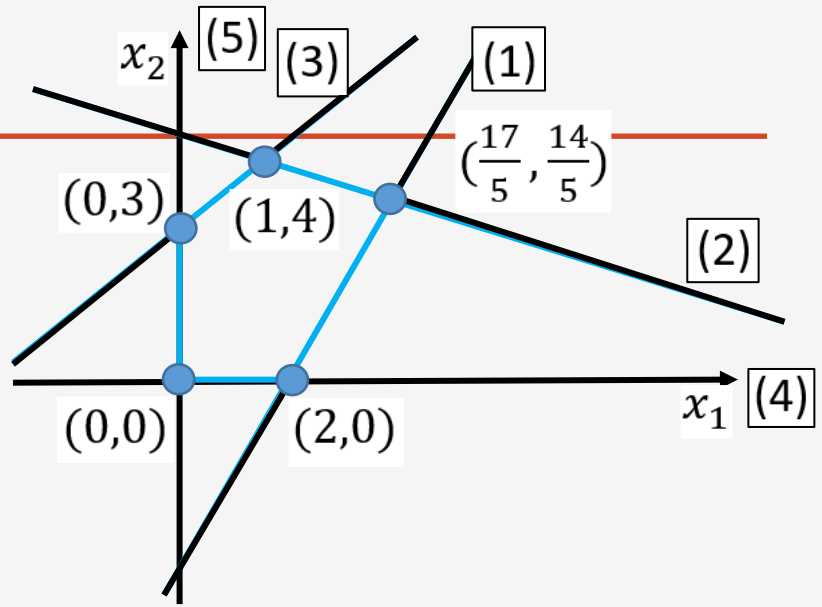


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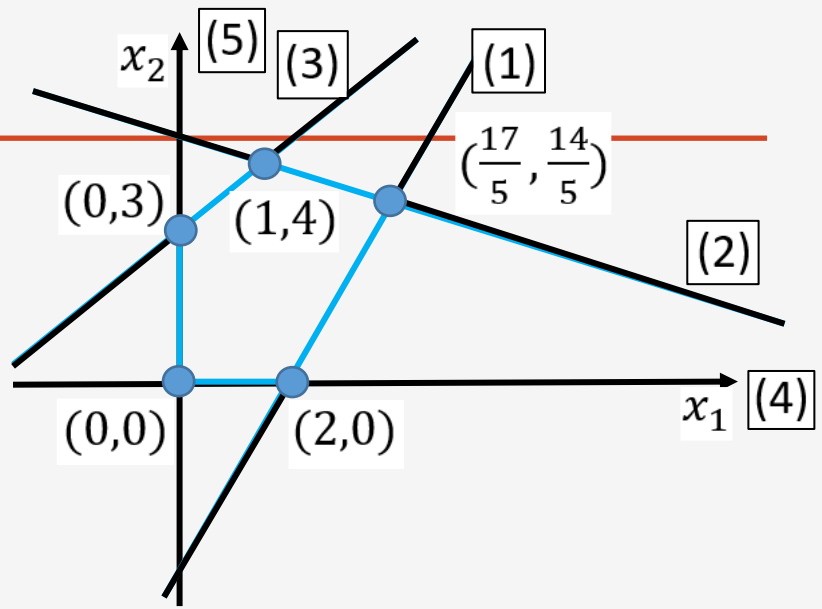


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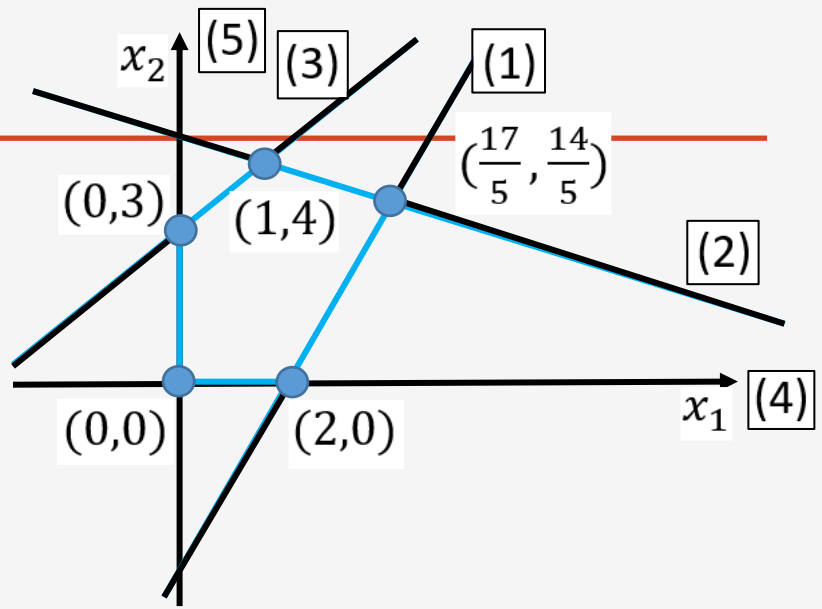


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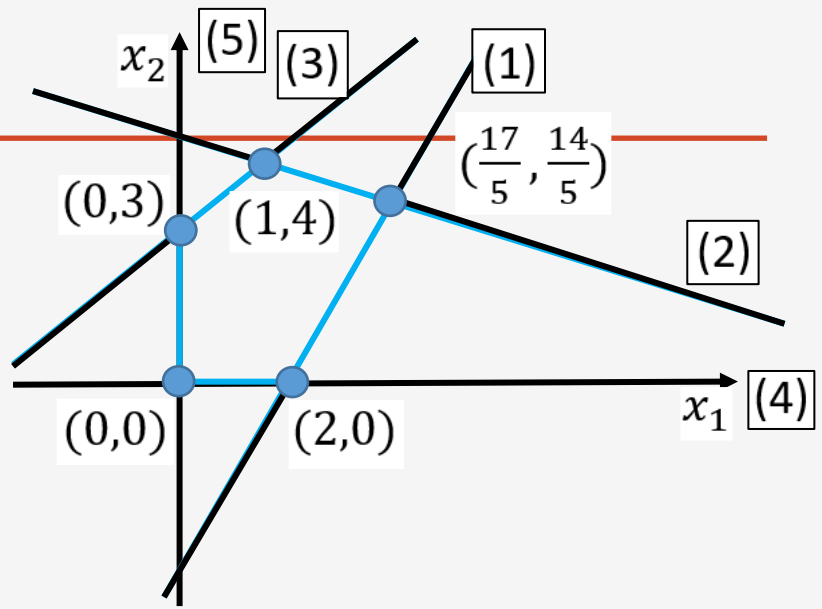


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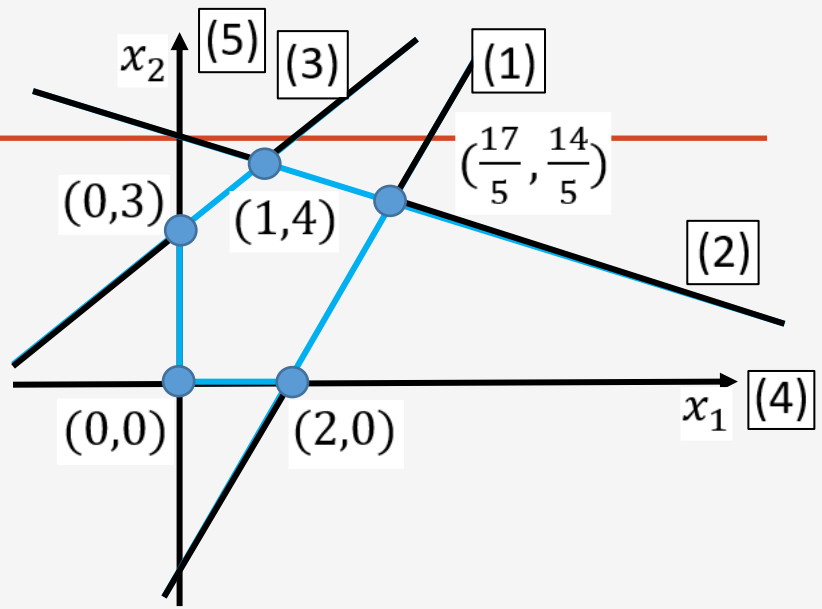


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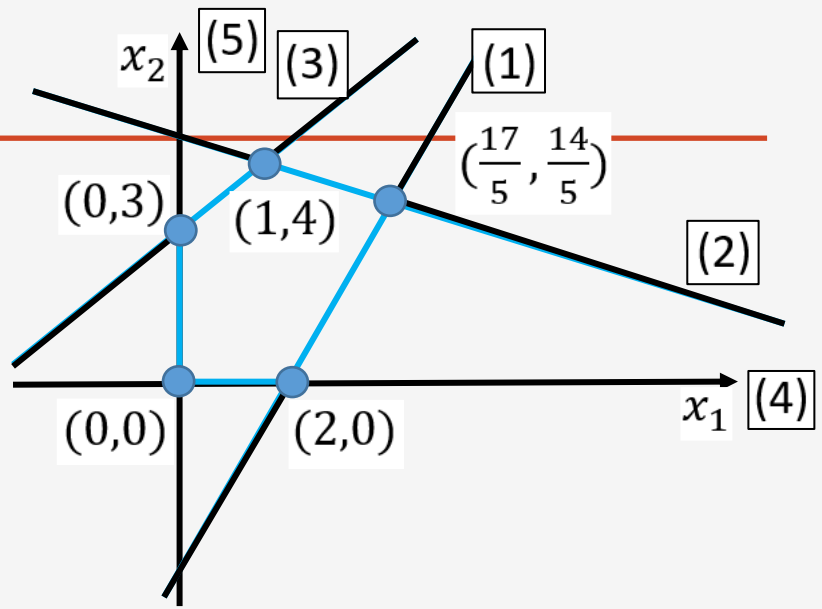


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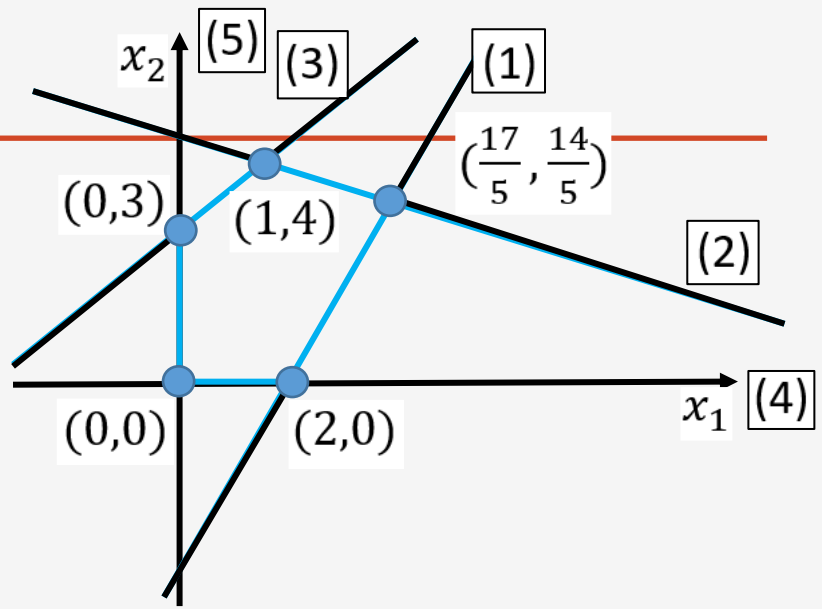


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- These are the tight constraints (3) and (4)

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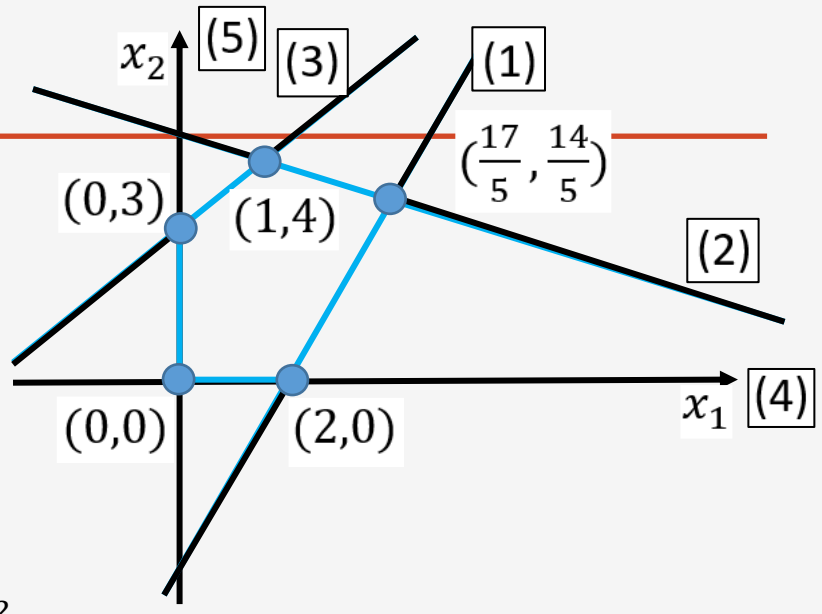


- For our example:
  - constraint (3) is now tight, so we can set $y_2 = 3 - (-x_1 + x_2) = 3 + x_1 - x_2$
  - constraint (4) is still tight so we can set $y_1 = x_1$
- By solving for $x_2$ and $x_1$ in the above two equations we have:

$$x_2 = 3 + y_1 - y_2$$
$$x_1 = y_1$$

- And substituting to the original LP we get an LP in terms of $(y_1, y_2)$ → let's call it LP(y)

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\max_{y \in R^2} \quad & 15 + 7y_1 - 5y_2,\\
s.t. \quad & y_1 + y_2 \le 7 && (1)\\
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Check if we can stop! Did we arrive at an optimal solution??

# The Simplex algorithm for LP

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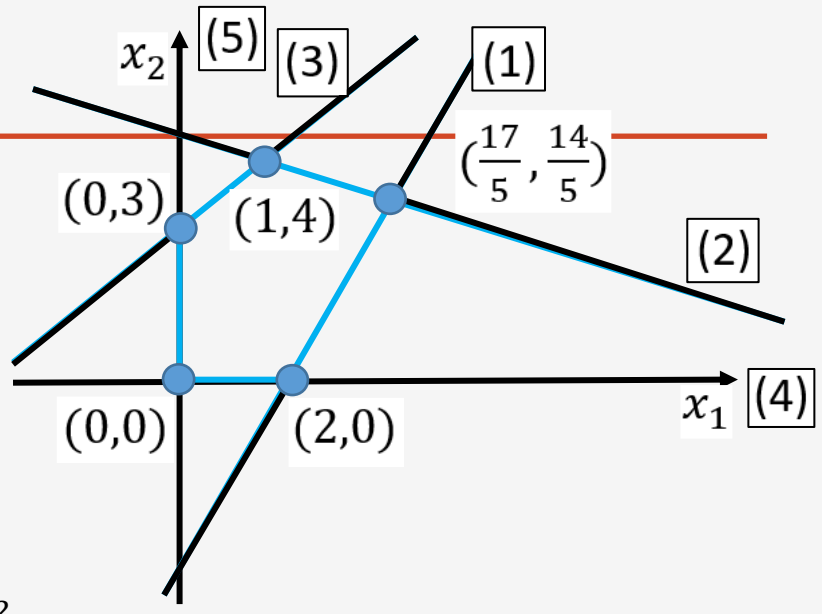


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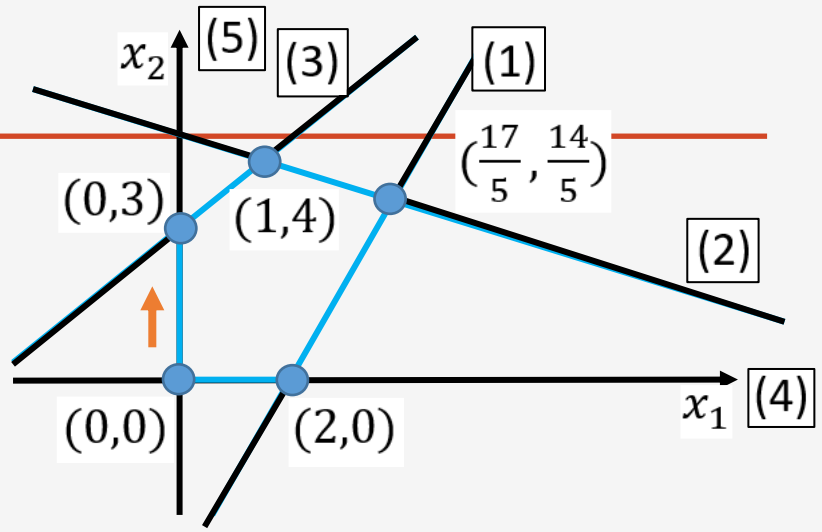
Check if we can stop! Did we arrive at an optimal solution??

** The origin $(y_1, y_2) = (0,0)$ is a vertex

** Since $c_1 = 7 > 0$, the origin is not Optimal, and we must continue

# The Simplex algorithm for LP




Summary of the main steps of **iter. #1:**

- Current vertex: $(x_1, x_2) = (0,0)$
- Tight constraints: {(4), (5)}
- Objective value = 0
- Select a $c_i$ with value $> 0$ → $c_2 = 5 > 0$
- **The Move:** increase $x_2$ and keep $x_1 = 0$
  - Stop when $x_2 = 3$ and constraint (3) becomes tight
- New tight constraints: {(4), (3)}
- New vertex: (0,3)
- Change coordinates:
  - $y_1 = x_1$
  - $y_2 = 3 + x_1 - x_2$
- Solve for $x_1 and\ x_2$
  - $x_1 = y_1$
  - $x_2 = 3 + y_1 - y_2$
- Substitute them to original LP to obtain new LP in $y_1 and\ y_2$
- Vertex at origin: $(y_1, y_2) = (0,0)$
- Check if all $c_i$'s are $< 0$
- If yes, Stop; Otherwise continue
  - $c_1 = 7 > 0$, so we continue…

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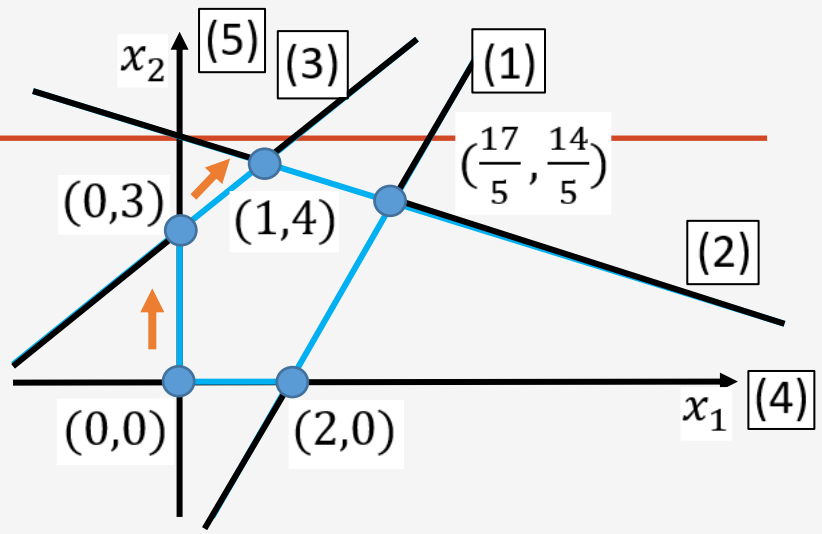
# The Simplex algorithm for LP




Summary of the main steps of **iter. #2:**

- Current vertex: $(y_1, y_2) = (0,0)$
- Tight constraints: {(4), (3)}
- Objective value = 15
- Select a $c_i$ with value $> 0$ → $c_1 = 7 > 0$
- **The Move:** increase $y_1$ and keep $y_2 = 0$
  - Stop when $y_1 = 1$ and constraint (2) becomes tight
- New tight constraints: {(2), (3)}
- New vertex: (1,4)
- Change coordinates:
  - $z_1 = 3 - 3y_1 + 2y_2$
  - $z_2 = y_2$
- Solve for $y_1$ *and* $y_2$
  - $y_1 = 1 - \frac{1}{3}z_1 + \frac{2}{3}z_2$
  - $y_2 = z_2$
- Substitute them to LP(y) to obtain new LP in $z_1$ *and* $z_2$
- Vertex is now at origin: $(z_1, z_2) = (0,0)$
- Check if all $c_i$'s are $< 0$
- YES $c_1 = -\frac{7}{3}$ and $c_2 = -\frac{1}{3}$ → Stop
  - $(z_1, z_2) = (0,0)$ is **optimal solution** of LP(z), $\text{Obj. Val}(0,0) = 22$

$$\begin{aligned} \max_{y \in R^2} \; & 15 + 7y_1 - 5y_2, \\ s.t. \quad & y_1 + y_2 \le 7 && (1) \\ & 3y_1 - 2y_2 \le 3 && (2) \\ & y_2 \ge 0 && (3) \\ & y_1 \ge 0\,, && (4) \\ & -y_1 + y_2 \le 3 && (5) \end{aligned}$$

LP(y)

→

$$\begin{aligned} \max_{z \in R^2} \; & 22 - \frac{7}{3}z_1 - \frac{1}{3}z_2, \\ s.t. \quad & -\frac{1}{3}z_1 + \frac{5}{3}z_2 \le 6 && (1) \\ & z_1 \ge 0 && (2) \\ & z_2 \ge 0 && (3) \\ & \frac{1}{3}z_1 - \frac{2}{3}z_2 \le 1\,, && (4) \\ & \frac{1}{3}z_1 + \frac{1}{3}z_2 \le 4 && (5) \end{aligned}$$

LP(z)

# The Simplex algorithm for LP

**BUT HOW DO WE RECOVER THE OPTIMAL SOLUTION OF THE ORIGINAL LP → $(x_1, x_2)$ ??**

- What we have achieved so far is to compute the optimal vertex (0,0) and the two tight constraints that define it in **LP(z)**

  $z_1 \geq 0 \quad (2)$
  $z_2 \geq 0 \quad (3)$

- **These constraints are also tight in the original LP (expressed in $(x_1, x_2)$).**
- Hence, we can solve the system of equations (taken from the original LP):

  $x_1 + 2x_2 = 9 \quad (2)$
  $-x_1 + x_2 = 3 \quad (3)$

- The solution is of the system of linear equations is: $(x_1, x_2) = (1,4)$.

- The optimal value of the **original LP** is **ObjVal(1,4)**$=2 \times 1 + 5 \times 4 = 22$

- **Note that 22 is also the optimal value of LP(z) for $(z_1, z_2) = (0, 0)$.**

# The Simplex algorithm for LP

We could have followed the path

$$(0,0) \rightarrow (2,0) \rightarrow \left(\frac{17}{5}, \frac{14}{5}\right) \rightarrow (1,4)$$

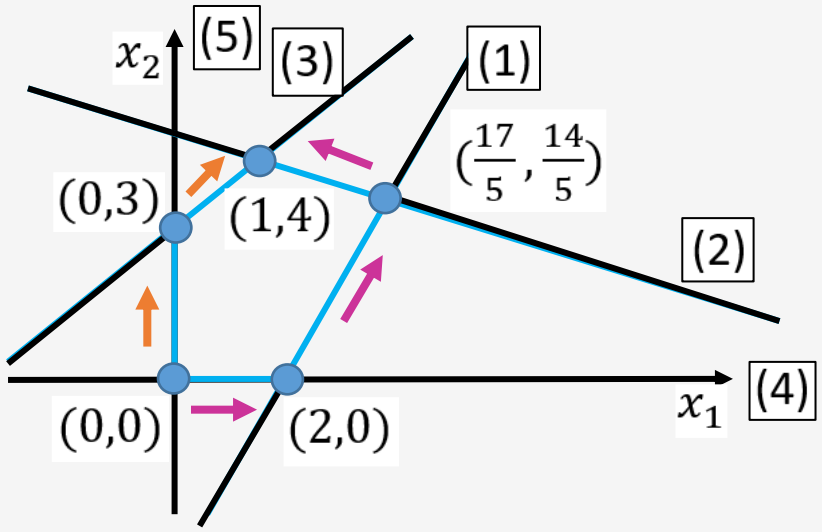
As you can see this is a longer path that the one we followed earlier.
It consists of one extra step.

Please use the previous slides to generate the purple path as an exercise (very good practice).

**Question:** Can you see why choosing (2,0) (instead of (0,3) lead to a longer path to the optimal solution (vertex)??

**Hint:** check the coefficients of the objective function

A possible answer is in the next slide →




$$\max_{x \in R^2} \; 2x_1 + 5x_2,$$

$$\begin{array}{lll} s.t. & 2x_1 - x_2 \leq 4 & (1) \\ & x_1 + 2x_2 \leq 9 & (2) \\ & -x_1 + x_2 \leq 3 & (3) \\ & x_1 \geq 0\,, & (4) \\ & x_2 \geq 0. & (5) \end{array}$$

# The Simplex algorithm for LP

We could have followed the path

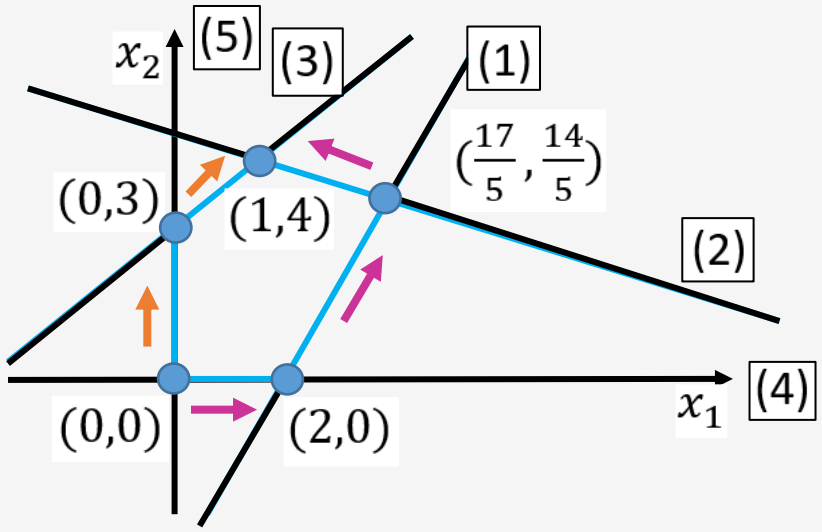
$$(0,0) \to (2,0) \to \left(\frac{17}{5}, \frac{14}{5}\right) \to (1,4)$$

As you can see this is a longer path that the one we followed earlier.
It consists of one extra step.

Please use the previous slides to generate the purple path as an exercise (very good practice).

**Question:** Can you see why choosing (2,0) (instead of (0,3) lead to a longer path to the optimal solution (vertex)??

**Answer:** Note that $c_2 = 5 > 2 = c_1$. Since we are maximizing, deciding to release constraint (5) and allow $x_2$ to take positive values results in larger values for the objective than releasing constraint (4) and allowing $x_1$ to take positive values.




$$\begin{aligned} \max_{x \in R^2} \; & 2x_1 + 5x_2, \\ s.t. \quad & 2x_1 - x_2 \le 4 && (1) \\ & x_1 + 2x_2 \le 9 && (2) \\ & -x_1 + x_2 \le 3 && (3) \\ & x_1 \ge 0\,, && (4) \\ & x_2 \ge 0. && (5) \end{aligned}$$